

XiangShan: An Open-Source High-Performance RISC-V Processor and Infrastructure for Architecture Research

The XiangShan Team
HPCA 26 @ Sydney, Australia
January 31, 2026

Opening Remarks

The Era of Open-Source Chip



A chip design that changes everything

- **10 breakthrough technologies 2023**

Ever wonder how your smartphone connects to your Bluetooth speaker, given they were made by different companies? Well, Bluetooth is an open standard, meaning its design specifications, such as the required frequency and its data encoding protocols, are publicly available. Software and hardware based on open standards—Ethernet, Wi-Fi, PDF—have become household names.

Now an open standard known as RISC-V (pronounced “risk five”) could change how companies create computer chips.

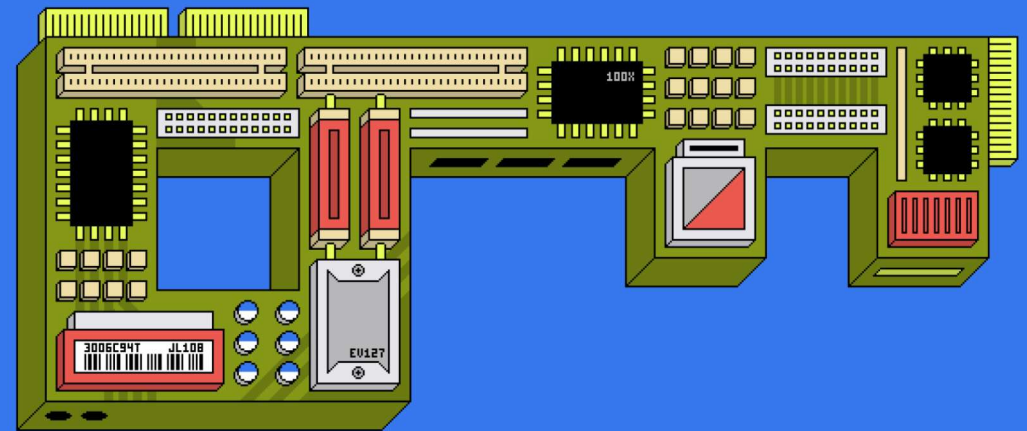
--- MIT Technology Review

A chip design that changes everything: 10 Breakthrough Technologies 2023

Computer chip designs are expensive and hard to license. That's all about to change thanks to the popular open standard known as RISC-V.

By Sophia Chen

January 9, 2023





L2: OPEN Design/Implementation

L3: OPEN Framework/Tools



Open Framework

microarch.

ISA Spec.

Open Flow

coding

Docs



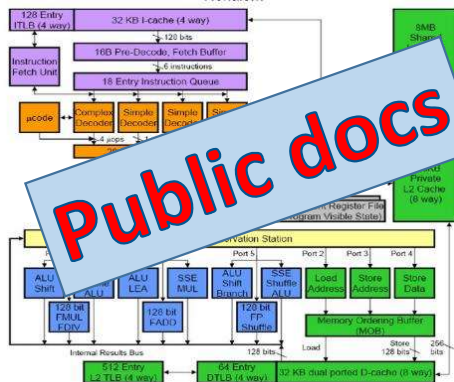
Open Tools

EDA Tools

RTL

Layout

Open & Free



```

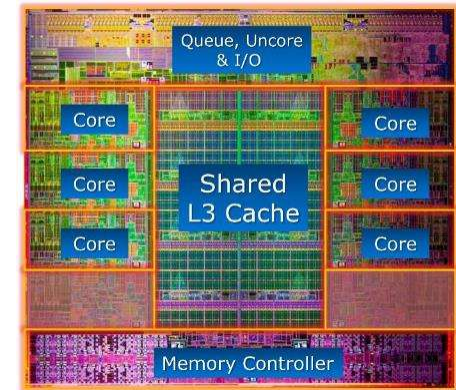
component DebugCoreTop is
port (
    -- Trigger and Data
    cu_Clk      : in    std_logic_vector(15 downto 0);
    cu0_Trig    : in    t_trig_0 := 0;
    cu1_Trig    : in    t_trig_1 := 0;
    cu2_Trig    : in    t_trig_2 := 0;
    cu0_Data    : in    std_logic_vector(15 downto 0);
    cu1_Data    : in    std_logic_vector(15 downto 0);
    cu2_Data    : in    std_logic_vector(15 downto 0);

    cu0_Logic   : in    std_logic := '0';
    cu1_Logic   : in    std_logic := '0';
    cu2_Logic   : in    std_logic := '0';

    gt_RX_p     : in    std_logic := '0';
    gt_RefClk_n : in    std_logic := '0';
    gt_RX_n     : in    std_logic := '0';
    gt_TX_p     : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_RX_n     : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_TX_p     : out   std_logic_vector(2 downto 0);
    gt_TX_n     : out   std_logic_vector(2 downto 0)
);
end component;

```

Open-source





Three levels of OSCE

L1: OPEN ISA

L2: OPEN Design & Implementation

L3: OPEN Tools & Infrastructure

3

Open Tools & Infrastructure

Microarch.

Engineering

EDA Tools

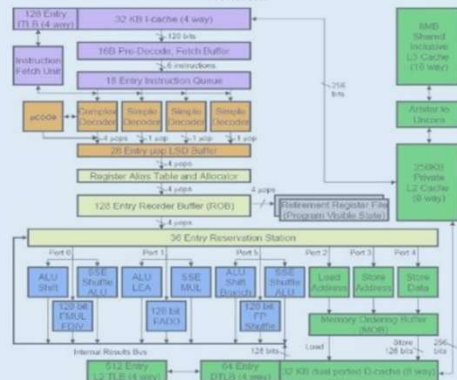
ISA Spec.



1

Open ISA

Docs



2

Open Design & Implt

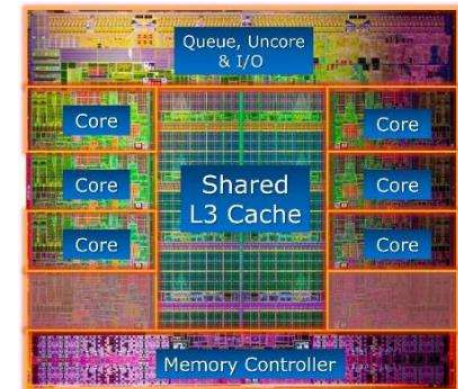
RTL codes

```
component DebugCoreTop is
  port (
    -- Trigger and Data
    cu_Clk      : in    std_logic_vector(2 downto 0) := (others => '0');
    cu0_Trig    : in    t_trig_0 := (others => (others => '0'));
    cu1_Trig    : in    t_trig_1 := (others => (others => '0'));
    cu2_Trig    : in    t_trig_2 := (others => (others => '0'));
    cu0_Data    : in    t_data_0 := (others => (others => '0'));
    cu1_Data    : in    t_data_1 := (others => (others => '0'));
    cu2_Data    : in    t_data_2 := (others => (others => '0'));

    -- Downstream I2C
    SCL         : in    std_logic := '0';
    SDA         : inout std_logic := '0';

    -- Upstream
    gt_RefClk_p : in    std_logic := '0';
    gt_RefClk_n : in    std_logic := '0';
    gt_RX_p     : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_RX_n     : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_TX_p     : out   std_logic_vector(2 downto 0);
    gt_TX_n     : out   std_logic_vector(2 downto 0);
  );
end component;
```

Layout





XiangShan: Open-Source High Performance Processors



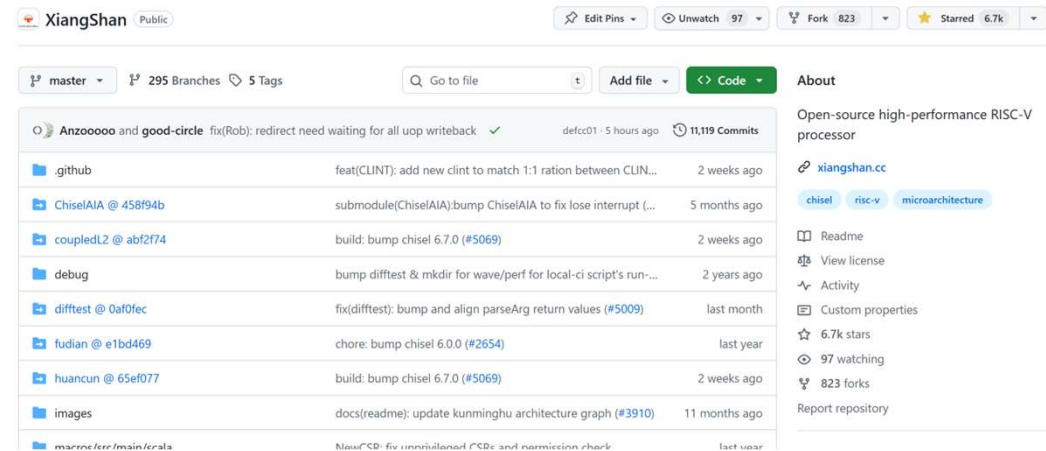
L1: OPEN ISA

L2: OPEN Design/Implementation

L3: OPEN Framework/Tools



Fragrant Hill in Beijing



> 6.8K stars, > 800 forks on GitHub

Why Open-Source High-perf. RISC-V Processor XiangShan?

- **Why RISC-V:** Free and open ISA
- **Why high-perf :** Most RISC-V processors are for IoT/edge, but both academic and industrial community need high-performance RISC-V processors
- **Why open-source:** An open and innovative hardware platform
- **Build a leading platform with end-to-end agile development flows and tools**



Linux

V.S



XIANGSHAN

**Envision:
“Hardware Version of Linux”**

Empower Architecture Research

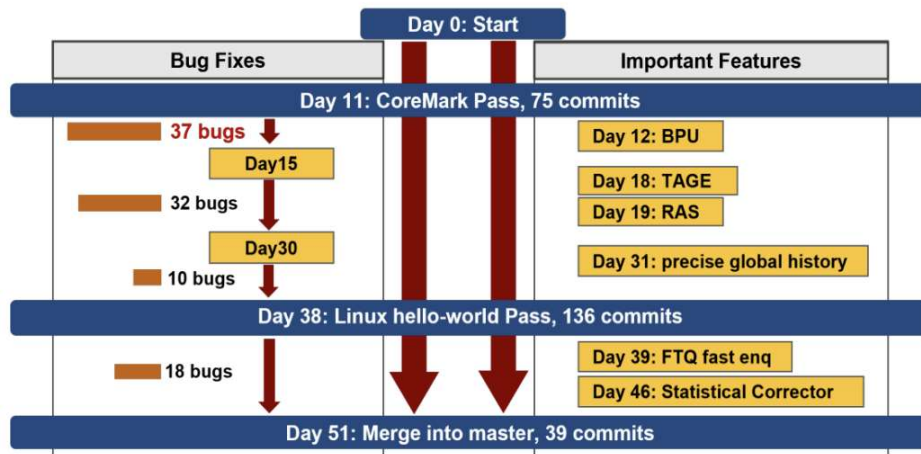
- An open-source, continuously developing research platform

Your research field	XiangShan provides ...
Microarchitecture	Performance: An industrial-competitive, high-performance superscalar OoO microarchitecture
	Functionality: RVA23-compatible RISC-V design
	Development: mature, user-friendly design flows
	Tapeouts by the XiangShan team and leading industrial partners with real-world deployment
Modeling	XS-GEM5: Gem5 simulator calibrated with XS-RTL
Chip development tool	Realistic and challenging research problems



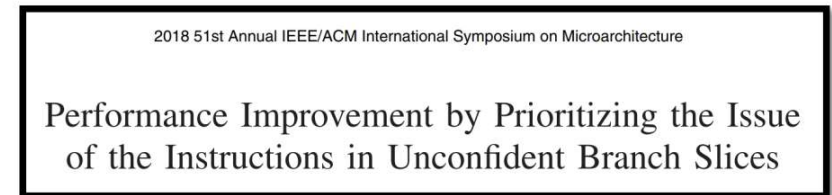
An Effective Infrastructure for Research

① Micro-architecture Optimization

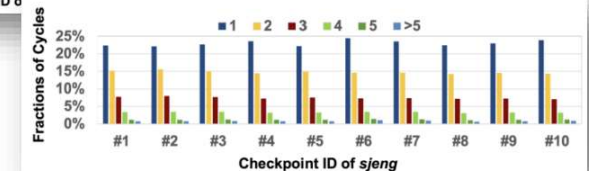
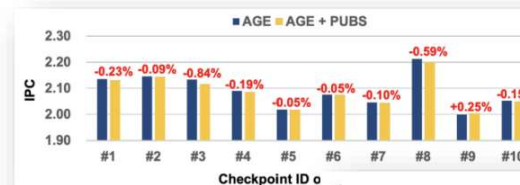


- 3 graduate students
- 11 days for a functionally correct prototype
- 37 bugs in 5 days
- 38 days to boot Linux with BPU
- 51 days for the overall frontend architecture

② Paper Reproduction



One third-year PhD student, 200 minutes on XiangShan





An Effective Infrastructure for Research

- **Topic: Computer Architecture**

- XiangShan: a realistic out-of-order RISC-V implementation with industry-competitive performance and an active open-source community
- MinJie provides the toolchains
- *Microarchitecture, accelerators, novel architectures, profiling, systems, benchmarking, security, compilers, ...*

- **Topic: Agile Chip Development**

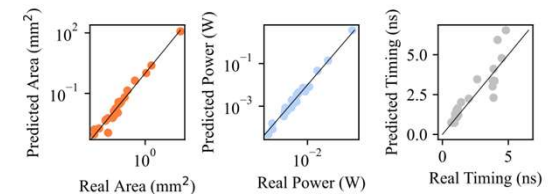
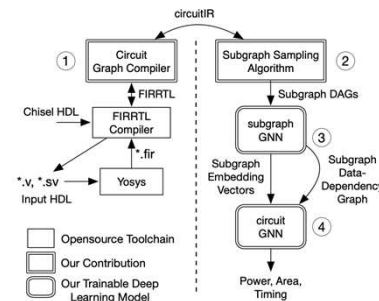
- XiangShan is a progressive, configurable, complicated, challenging benchmark
- MinJie provides a good startpoint
- *HDLs, verification, performance, power, area, prototyping, DFT, synthesis, placement, routing,*



midgard

EPFL

Imprecise Store Exceptions, ISCA'23
Single Address Space FaaS with Jord, ISCA'25



SNS v2, MICRO'23 (Duke University)

Research on Xiangshan

Work	Venue
DiffTest-H, You et al.	MICRO 2025
GSIM, Chen et al.	DAC 2025
Imprecise Store Exceptions, Gupta et al.	ISCA 2023
TEESec, Ghaniyoun et al.	ISCA 2023
Xu et al.	MICRO 2023
Khronos, Zhou et al.	MICRO 2023
Wang et al.	ICCAD 2023
REMU, Chen et al.	ICCD 2023
Wang et al.	TACO
Single-Address-Space FaaS with Jord, Li et al.	ISCA 2025
CoroAMU, Jiang et al.	PACT 2025
.....	



15+ XiangShan Tutorials Around the World

- HPCA, Edinburgh, Scotland
- ASPLOS, San Diego, USA
- RVSC, Hangzhou, China
- MICRO, Austin, USA

2024

- **HPCA, Sydney, Australia**
- More coming ...

2026

2023

- ASPLOS, Vancouver, Canada
- RVSC, Beijing, China
- MICRO, Toronto, Canada

2025

- HPCA, Las Vegas, USA
- ASPLOS, Rotterdam, Netherlands
- RVSE, Paris, France
- ISCA, Tokyo, Japan
- APPT, Athens, Greece
- RVSC, Shanghai, China
- MICRO, Seoul, Korea

***Welcome old
and new friends!***



What We Will Cover in This Tutorial



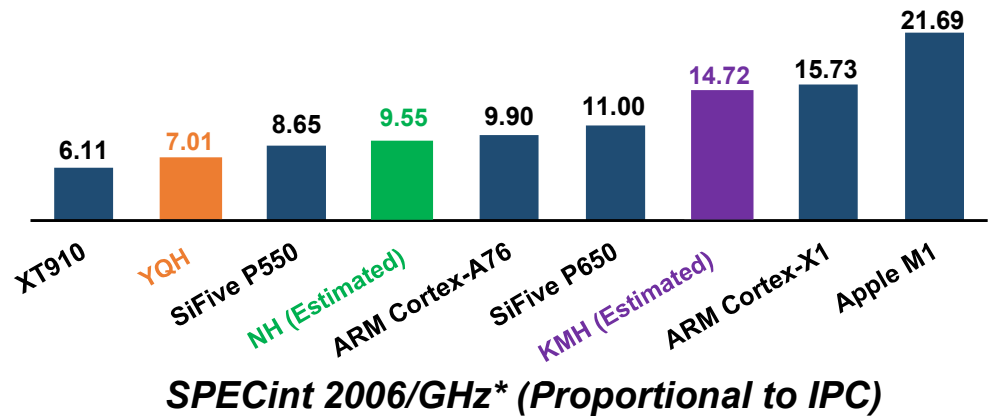
- **08:45 Introduction**
- 09:15 Microarchitecture Design Philosophy of XiangShan
- 10:10 XSCC: A High-Performance Compiler for RISC-V
- 10:20 Baiyang: High Performance Open-Source DDR Controller IP
- 10:30 Coffee Break
- 11:00 Open-Source Tools and Open Problems in Agile Chip Development Infrastructure
- 11:15 XS-GEM5 and Hands-on
- 11:55 Development Workflows and Hands-on

Open-Source, High Performance, Industrial Level RISC-V Processor



XiangShan: Open-Source High Performance Processors

- **1st generation: Yanqihu (YQH)**
 - RV64GC, single-core, superscalar OoO
 - 28nm tape-out, 1.3GHz, July 2021
 - SPEC CPU2006 7.01@1GHz, DDR4-1600
- **2nd generation: Nanhu (NH)**
 - RV64GCBK, dual-core, superscalar OoO
 - 14nm GDSII delivery, 2GHz, 2023 Q3
 - Estimated** SPECint 2006 19.10@2GHz
- **3rd generation: Kunminghu (KMH)**
 - RV64GCBKHV, quad-core, superscalar OoO
 - Advanced-node, 3GHz, 1.5x IPC of NH
 - Close collaboration with industrial partners



* Source: XT910@ISCA'20, SiFive, AnandTech

** Updated January 5, 2023

XiangShan Gen 1: Yanqihu

- **Test chip developed almost entirely by students**
 - RV64GC, 11-stage, superscalar, out-of-order
 - 5.3 CoreMark/MHz (gcc-9.3.0 -O2)
- **Tape-out:** single XiangShan core with 1MB L2 Cache
 - 28nm, 1.3GHz, July 2021



Yanqi Lake in Beijing

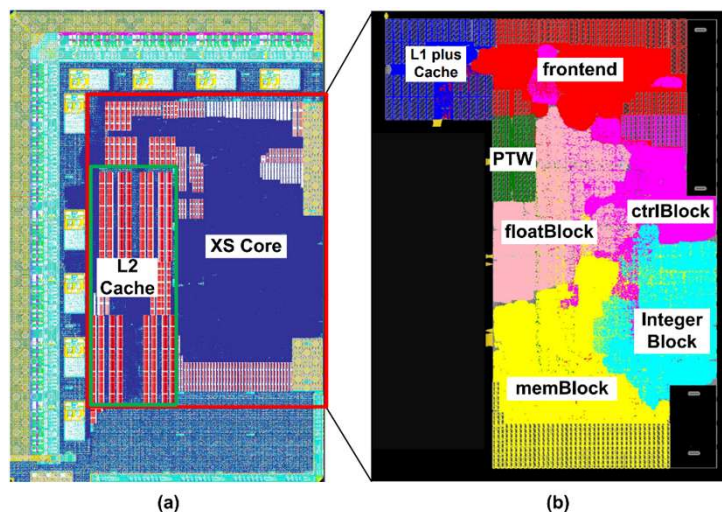


Figure. Layout of (a) the entire chip; (b) the core

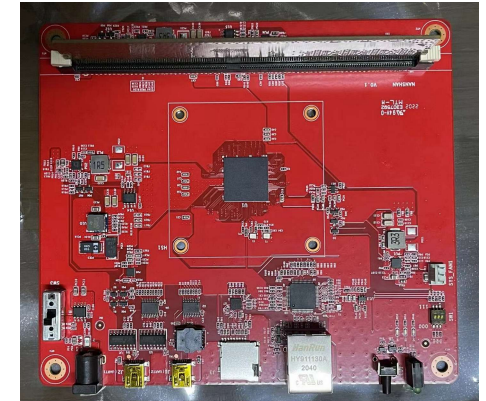


Tape-out information for the processor core	
Process Node	28nm
Die Size	8.6 mm ²
Std Cell	5.05M, 4.27 mm ²
Mem	261, 1.7mm ²
Density	66%
Cell	ULVT 1.04%, LVT 19.32%, SVT 25.19%, HVT 53.67%
Estimated Power	5W
Frequency	1.3GHz, TT85C



Real Chip of XiangShan Gen 1 Yanqihu

- The chip was back in January 2022
 - SoC: CPU, SPI Flash, UART, SD card, Ethernet, DIMM
 - Correctly running Debian with SD card and ethernet
- Performance: SPEC CPU2006 7.01@1GHz



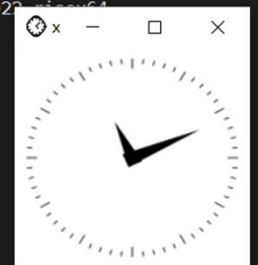
SPECint 2006 @ 1GHz		SPECfp 2006 @ 1GHz	
400.perlbench	6.14	410.bwaves	9.28
401.bzip2	4.37	416.gamess	6.59
403.gcc	6.71	433.milc	8.41
429.mcf	6.83	434.zeusmp	7.65
445.gobmk	7.92	435.gromacs	4.99
456.hmmer	5.24	436.cactusADM	3.97
458.sjeng	6.85	437.leslie3d	6.93
462.libquantum	17.71	444.namd	8.00
464.h264ref	10.91	447.dealII	10.17
471.omnetpp	5.65	450.soplex	7.03
473.astar	5.16	453.povray	7.14
483.xalancbmk	7.35	454.Calculix	2.86
		459.GemsFDTD	8.35
		465.tonto	6.42
		470.lbm	10.39
		481.wrf	7.26
		482.sphinx3	9.07

SPECint 2006: 7.03@1GHz
SPECfp 2006: 7.00@1GHz

```
wanghuizhe@open02:~$ ssh -X xs@172.28.2.246
xs@172.28.2.246's password:
Linux open02 4.20.0-44668-ge9c195ab0c63-dirty #109 Thu Feb 17 17:41:13 CST 2022; root@open02

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
You have no mail.
Last login: Thu Feb 17 11:10:31 2022 from 172.28.9.102
xs@open02:~$ xclock
Warning: locale not supported by C library, locale unchanged
```



**SSH into the Debian on
XiangShan, and
run a GUI program via X11
forwarding**

XiangShan Gen 2: Nanhu

- **Target: 2GHz@14nm, SPEC CPU2006 ~20 marks**
- **New frontend:** decoupled BP and instruction fetch
- **Improved backend:** better scheduler, instruction fusions, move elimination, and more
- **New L2/L3 cache:** designed for high frequency and high performance with hybrid prefetchers
- **Verified under dual-core config (RV64GCBK), more devices support (PCIe, USB, ...)**

Setup an open and standardized development workflow



A lake in Jiaying, Zhejiang, China

☐	 10 Open	☒ 2,322 Closed
☐	 Bump difftest for palladium simulation ✓	
	#2662 opened 11 hours ago by klin02 • Approved	
☐	 ICache: fix ICacheMainPipe bug about sfence ✓	bug do not merge
	#2660 opened 3 days ago by ssszwic • Review required	
☐	 ICache: change data SRAM partWayNum from 2 to 4 ✓	
	#2653 opened 5 days ago by ssszwic • Approved	
☐	 ITTAGE meta width shrink ✓	do not merge
	#2592 opened last month by eastonman • Review required	
☐	 pf: fix negative stream ✓	bug
	#2508 opened on Nov 27, 2023 by happy-lx • Review required	
☐	 SQ: add sq merge ✓	do not merge enhancement
	#2439 opened on Oct 29, 2023 by sfencevma • Review required	
☐	 rm refillPipe ✓	enhancement
	#2426 opened on Oct 25, 2023 by YukunXue • Approved	

Pull Request Snapshot

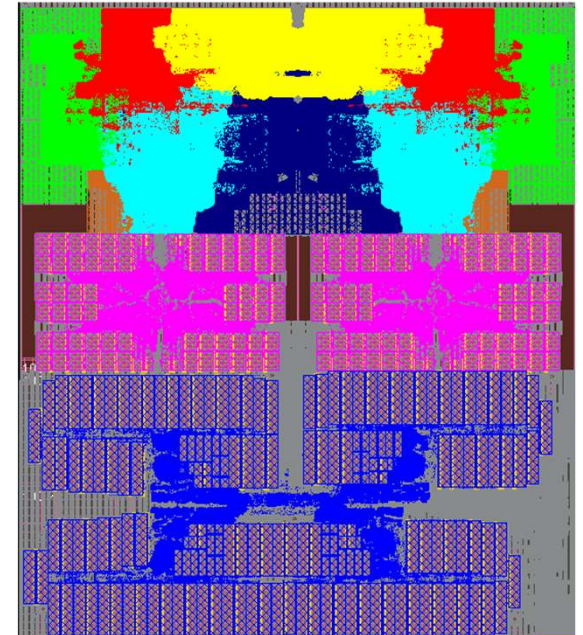


Estimated Performance of XiangShan Gen 2 Nanhu

- Estimated **SPECint 2006 19.10, SPECfp 2006 22.18@2GHz**
 - RTL simulation, DDR4-2400 under DRAMsim3
 - Compile with GCC 10.2.0, -O2, RV64GCB

SPECint 06	
400.perlbench	19.27
401.bzip2	11.36
403.gcc	21.97
429.mcf	20.53
445.gobmk	15.97
456.hmmer	19.22
458.sjeng	17.22
462.libquantum	36.99
464.h264ref	28.54
471.omnetpp	14.02
473.astar	14.19
483.xalancbmk	21.48
SPECint2006@2GHz	19.10

SPECfp 06	
410.bwaves	18.09
416.gamess	23.82
433.milc	18.33
434.zeusmp	28.19
435.gromacs	17.53
436.cactusADM	24.26
437.leslie3d	20.28
444.namd	23.83
447.dealII	33.50
450.soplex	25.61
453.povray	27.06
454.Calculix	9.18
459.GemsFDTD	24.66
465.tonto	17.68
470.lbm	32.04
481.wrf	19.73
482.sphinx3	28.38
SPECfp2006@2GHz	22.18



Feature	V1 YQH	V2 NH
ISA	RV64GC	RV64GCBK
Process Node	28nm	14nm
Core Count	1	2
Die Size	8.6 mm ²	22.13 mm ²
Std Cell Num/Area	5.05M, 4.27 mm ²	11.3M, 4.53 mm ²
Mem Num/Area	261, 1.7 mm ²	692, 8.93 mm ²
Density	66%	35%
Frequency	1.3GHz, TT 0.9V	2GHz, TT 0.9V

* Updated January 5, 2023. Performance confirmed on FPGA as well

XiangShan Gen 3: Kunminghu

- **Target ARM Neoverse N2**
 - SPECCPU2006: 45@3GHz (15/GHz)
 - Vector/Hypervisor extension supported
- **A Joint Dev Team** (coordinated by BOSC)
 - About 10 institutions



Kunming Lake in Beijing





Highlights in XiangShan Gen 3 Kunminghu

- **Functional Enhancement**

- Support RISC-V **Vector/Hypervisor** extension
- Support **RVA23 profile**
- Support interconnection based on **CHI protocol**

- **Performance Exploration**

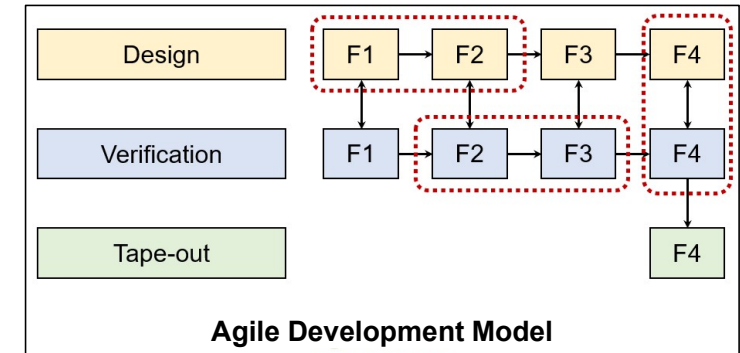
- Performance boost in frontend, backend, load-store unit and cache
- Performance model calibrated with RTL
- Workflow: **DSE on perf model => Impl. & fine tuning on RTL**

- **Functional Verification**

- **Hierarchical verification flow** spanning system/integration/unit level + FPGA prototyping
- Industrial-grade verification process

- **Physical Design**

- Experienced physical design team
- Simultaneous iteration of RTL coding based on timing evaluation





Compiler Support

- LLVM & GCC Now Include XiangShan Kunminghu V2R2 Optimizations

llvm / llvm-project

[Code](#) [Issues](#) 5k+ [Pull requests](#) 5k+ [Actions](#) [Security](#) [Insights](#)

Commit 0e3e0bf

4 people authored on Apr 21, 2025 · 21 / 23 · [Verified](#)

[RISC-V] Add processor definition for XiangShan-KunMingHu-V2R2 (#123193)

XiangShan-KunMingHu is the third generation of Open-source high-performance RISC-V processor developed by Beijing Institute of Open Source Chip (BOSC), and its latest version is V2R2.

The KunMingHu manual is now available at <https://github.com/OpenXiangShan/XiangShan-User-Guide/releases>. It will be updated on the official XiangShan documentation site: <https://docs.xiangshan.cc/zh-cn/latest>

You can find the corresponding ISA extension from the XiangShan Github repository: <https://github.com/OpenXiangShan/XiangShan/blob/master/src/main/scala/xiangshan/Parameters.scala>

If you want to track the latest performance data of KunMingHu, please check XiangShan Biweekly: <https://docs.xiangshan.cc/zh-cn/latest/blog>

This PR adds the processor definition for KunMingHu V2R2, developed by the XSCC team <https://github.com/orgs/OpenXiangShan/teams/xscc>.

The scheduling model for XiangShan-KunMingHu V2R2 will be submitted in a subsequent PR.

Co-authored-by: Shenglin Tang <tangshenglin@ict.ac.cn>
Co-authored-by: Xu, Zefan <ceba_robot@outlook.com>
Co-authored-by: Tang Haojin <tanghaojin@outlook.com>

main (#123193) · llvmorg-22-init ... llvmorg-21.1.0-rc1

gcc-mirror / gcc

[Code](#) [Pull requests](#) 49 [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#)

Commit f0cd40f

3 people committed on Jun 6, 2025

RISC-V: Support -mcpu for XiangShan Kunminghu cpu.

This patch adds support for the XiangShan Kunminghu CPU in GCC, allowing the use of the `-mcpu=xiangshan-kunminghu` option.

XiangShan-KunMingHu is the third-generation open-source high-performance RISC-V processor.[1] You can find the corresponding ISA extension from the XiangShan Github repository.[2] The latest news of KunMingHu can be found in the XiangShan Biweekly.[3]

[1] <https://github.com/OpenXiangShan/XiangShan-User-Guide/releases>.
[2] <https://github.com/OpenXiangShan/XiangShan/blob/master/src/main/scala/xiangshan/Parameters.scala>
[3] <https://docs.xiangshan.cc/zh-cn/latest/blog>

A dedicated scheduling model for KunMingHu's hybrid pipeline will be proposed in a subsequent PR.

gcc/ChangeLog:

```
* config/riscv/riscv-cores.def (RISCV_TUNE): New cpu tune.  
(RISCV_CORE): New cpu.  
* doc/invoke.texi: Ditto.
```

gcc/testsuite/ChangeLog:

```
* gcc.target/riscv/mcpu-xiangshan-kunminghu.c: New test.
```

Co-Authored-By: Jiawei Chen <jiawei@iscas.ac.cn>
Co-Authored-By: Yangyu Chen <cyy@cyyself.name>
Co-Authored-By: Tang Haojin <tanghaojin@outlook.com>

master

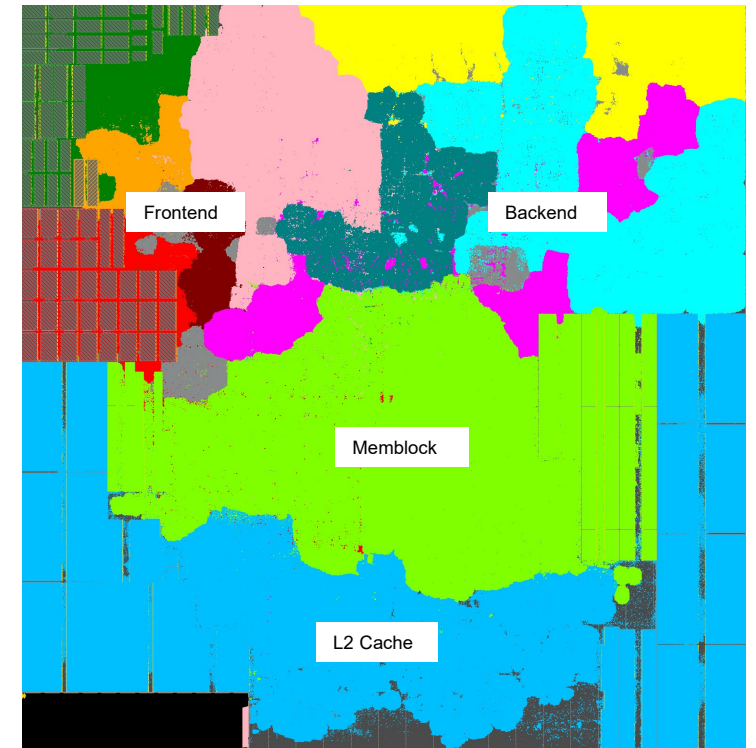


Performance Evaluation of Gen 3 Kunminghu

- Method: SPEC CPU checkpoints selected by Simpoint
 - Base: GCC 12 -O3, RV64GCB, jemalloc
 - 1MB L2 and 16MB L3
 - Simulated@3GHz with DRAMsim3 DDR4-3200

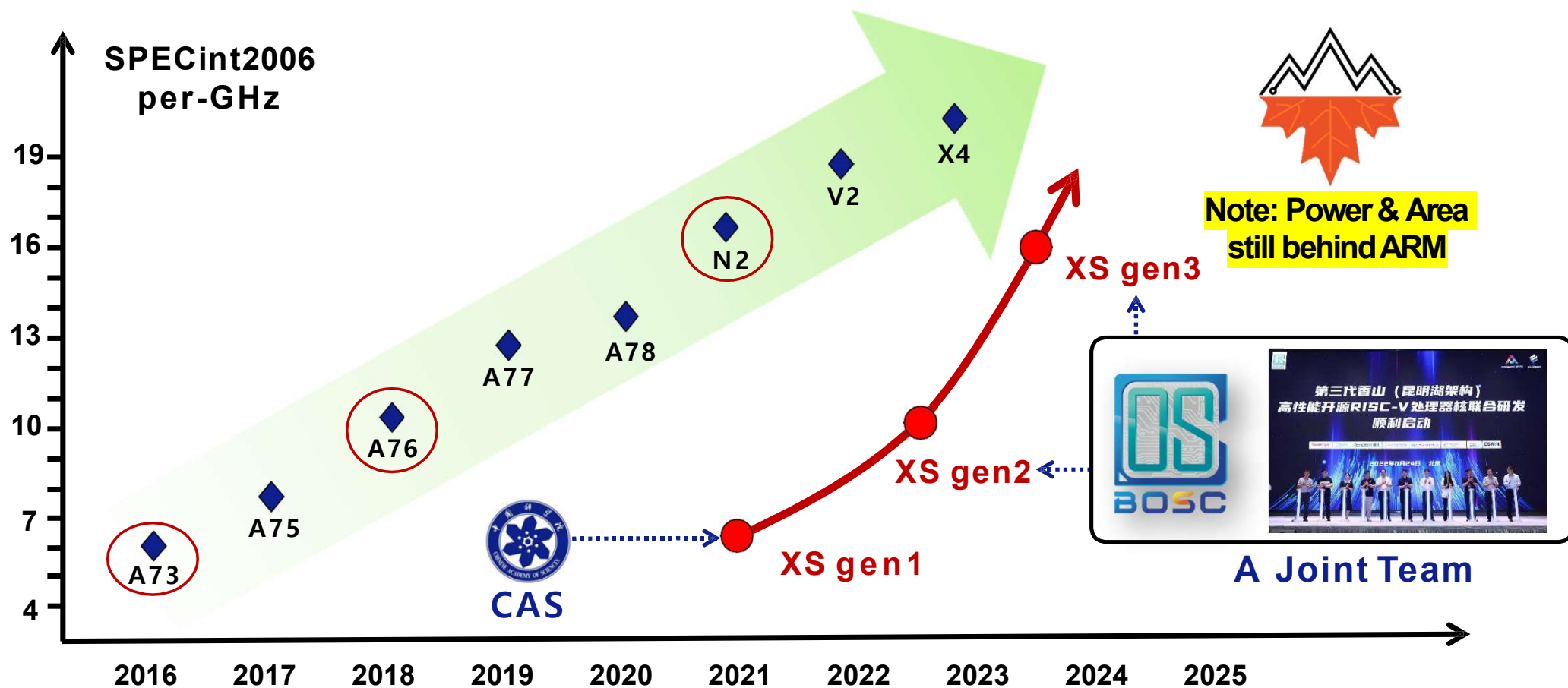
SPECint 2006 est.@ 3GHz		SPECfp 2006 est.@ 3GHz	
400.perlbench	35.88	410.bwaves	66.89
401.bzip2	25.50	416.gamess	40.89
403.gcc	46.72	433.milc	45.25
429.mcf	58.13	434.zeusmp	52.10
445.gobmk	30.26	435.gromacs	33.65
456.hmmer	41.60	436.cactusADM	46.16
458.sjeng	30.53	437.leslie3d	46.01
462.libquantum	122.50	444.namd	28.88
464.h264ref	56.57	447.dealII	73.43
471.omnetpp	39.37	450.soplex	51.99
473.astar	29.23	453.povray	53.44
483.xalancbmk	72.03	454.Calculix	16.38
GEOMEAN	44.15	459.GemsFDTD	37.18
		465.tonto	36.67
		470.lbm	91.24
		481.wrf	40.62
		482.sphinx3	48.57
		GEOMEAN	44.60

* Updated in March 2025



Floorplan of KMH V2R2 (single core)

XiangShan: Open-Source High Performance Processors



Prospect: "Dual Core" Roadmap



Based on **gen3**
(KMH)

- **Big Core: Ultimate Performance (v.s. ARM N2)**

Target High-Throughput Advanced Computing Platform

Goal : become mainstream CPU for data centers and computational facilities



Based on **gen2**
(NH)

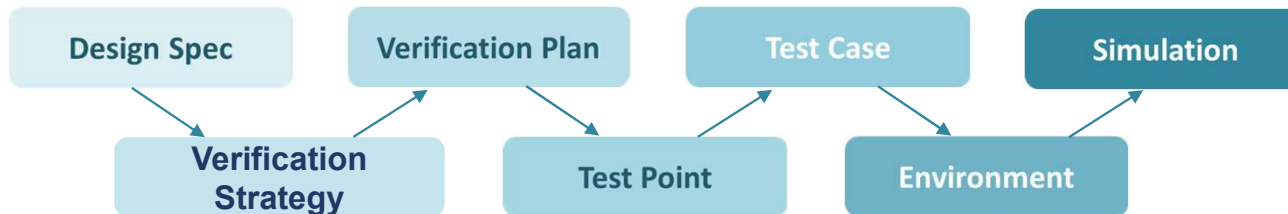
- **Mid Core: Balanced Perf & Efficiency (v.s. ARM A76)**

Target Mid-to-High-End General Industry Domain

Goal: support broad industrial spectrum including industrial control, automotive, communication, aviation and more

Industrial Level Verification

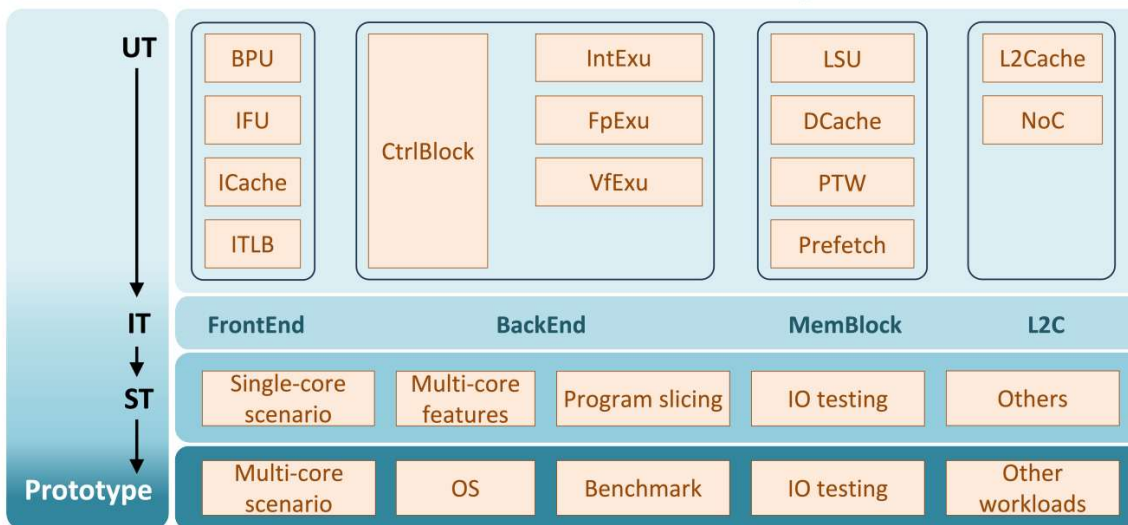
① Process



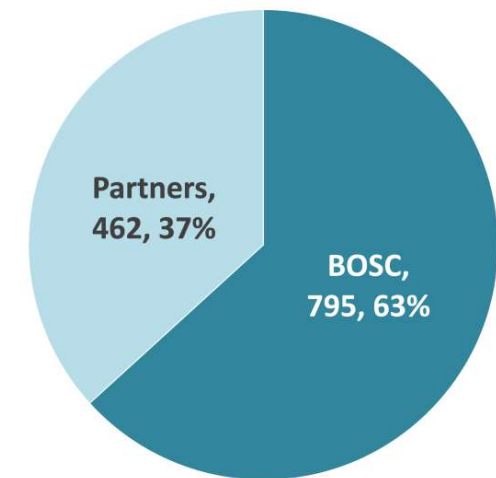
② Tools

UVM UnityChip XFuzz
SPIKE DiffTest FORCE-RISCV
RISCV-DV EMU FPGA SIM
EDA Business tools Formal

③ Verification Hierarchy



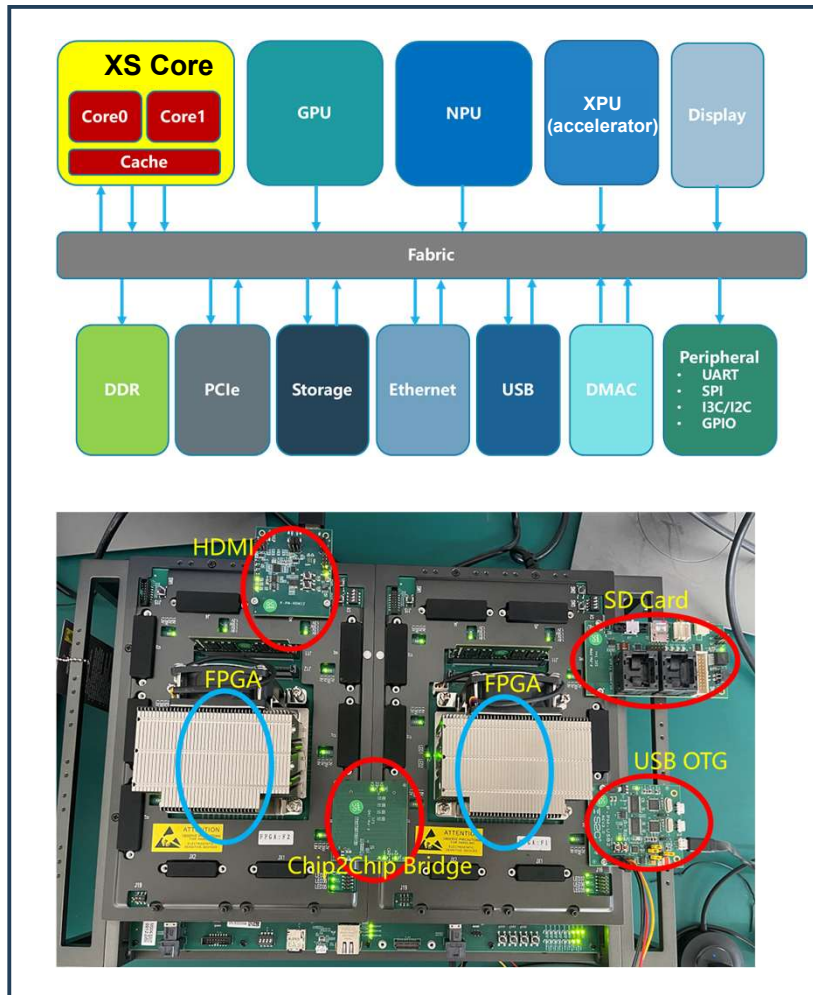
④ Partner Collaboration



Source and Number of Reported Bugs



FPGA Prototype in Only Two Weeks



Source: Xinchen Technology

🏔️ Real Chip of XiangShan Gen 2 Nanhu

- **Test chip was back in October 2023**
 - Successfully brought up Linux and working with PCIe device (GPU, Ethernet, USB..)



Nanhu test board @ RVSC'24
Running Desktop Linux & Cloud Game*

The world's first laptop powered by a open-source RISC-V processor

ISCAS
中国科学院软件研究所
Institute of Software, Chinese Academy of Sciences

Ruyi Book

CPU	"XiangShan Nanhu" (RV64GCBK), up to 2.5GHz
Memory	8GB DDR5 4800MT/s
GPU	AMD RX 550
USB	2x USB3
Ethernet	2x 2.5Gbps Ethernet Port
Display	1x 14-inch LCD Display 1x HDMI, up to 4K
TouchPad	Support 9 kinds of gesture operation
Audio	Built-in high-quality speakers.
Dimensions	315*233*25mm

Powered by

XIANGSHAN
"XiangShan" high-performance open-source RISC-V processor

Ruyi Book

inchi 英麒智能 milkv

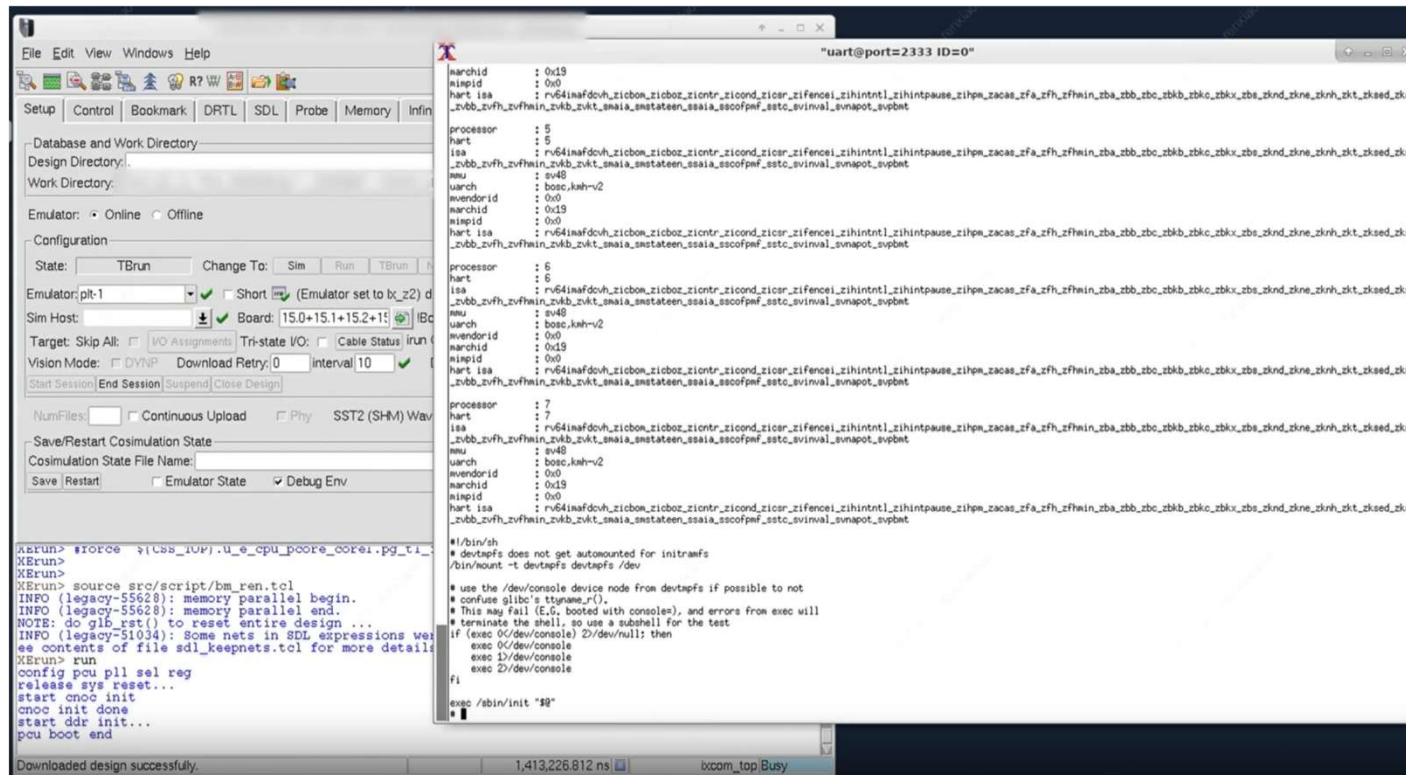
Ruyi XiangShan Book
Design by ISCAS, Inchi, MilkV

* Genshin Impact · Cloud, only ~ 1 fps, just for fun

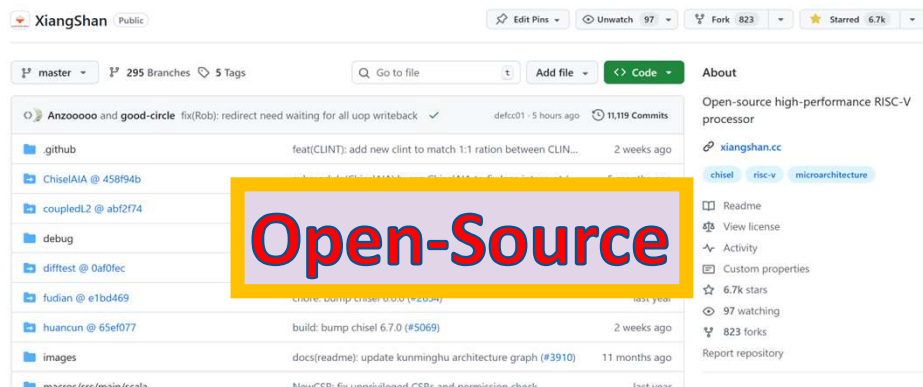


Linux Boot on 8-Core XiangShan Gen 3 Kunminghu V2

- Lanxin Computing has successfully launched Linux on an 8-core SoC built on the Kunminghu V2

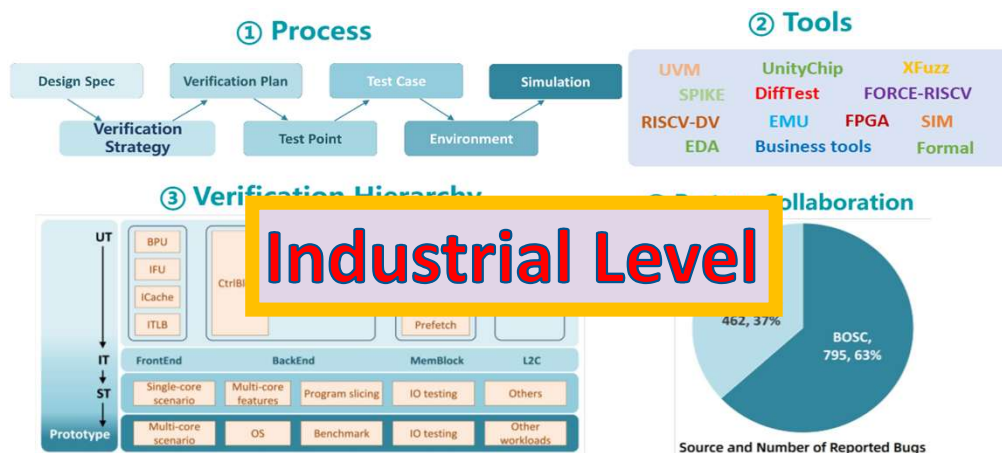
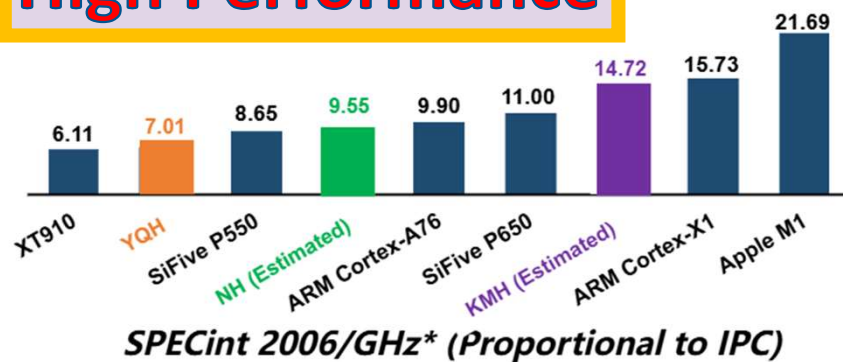


XiangShan: A RISC-V Processor



> 6.8K stars, > 850 forks on GitHub

High Performance



Industrial Level

User-Friendly Infrastructure



First Step to Agile Design: Use Chisel

• 2018: quantitative experiments between Chisel and Verilog

• **Task #1:** Design an L2 Cache for RISC-V Rocket-chip core

• **Who:** A 5-year engineer vs. a senior student

	A 5-year Engineer	An Undergraduate
Experience	Familiar w/ OpenSparc T1; Modified Xilinx Cache	Course projects on CPU design; Using Chisel for 9 months
Language	Verilog	Chisel
Time	6 weeks	3 days
LOCs	~1700	~350
Results	Unable to boot Linux	Boot Linux on multicore RISC-V; support NIC w/ DMA mode

• **1st Round results:** Chisel is more productive than Verilog by **14X** with only **1/5** LOC

• **Task #2:** Translate the Verilog codes into Chisel

• Evaluated on FPGA (xc7v2000tffg1716-1), Vivado 2017.01

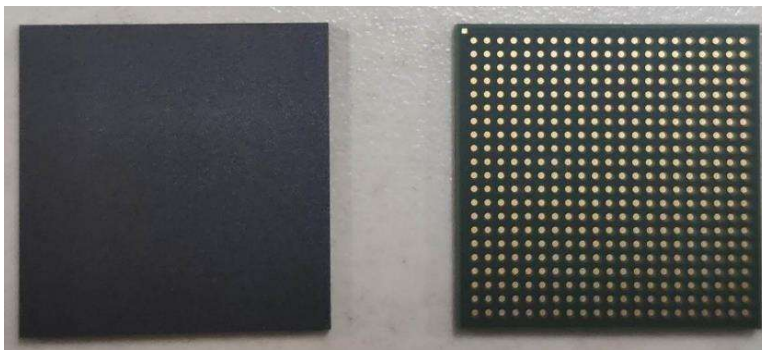
• **Who:** A junior student who never knew Chisel

	Verilog	Chisel (direct translation)	Chisel-opt (adv. features & libs)
Freq./MHz	135.814	136.388 (+0.42%)	154.107 (+13.47%)
Power/W	0.770	0.749 (-2.73%)	0.749 (-2.73%)
LUT Logic	5676	6422 (+13.14%)	2594 (-54.30%)
LUT Storage	1796	1264 (-29.62%)	1492 (-16.93%)
FF	4266	3638 (-14.72%)	747 (-82.49%)
LOCS	618	470 (-23.95%)	155 (-74.92%)

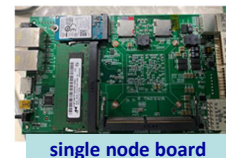
• **2nd Round results:** Chisel can achieve better PPA than verilog

Yu Zihao, Liu Zhigang, Li Yiwei, Huang Bowen, Wang Sa, Sun Ninghui, Bao Yungang. Practice of Chip Agile Development: Labeled RISC-V. Journal of Computer Research and Development, 2019, 56(1): 35-48.

• 2020: 28-nm tape-out of an 8-core labeled RISC-V processor



16-node prototype case



single node board

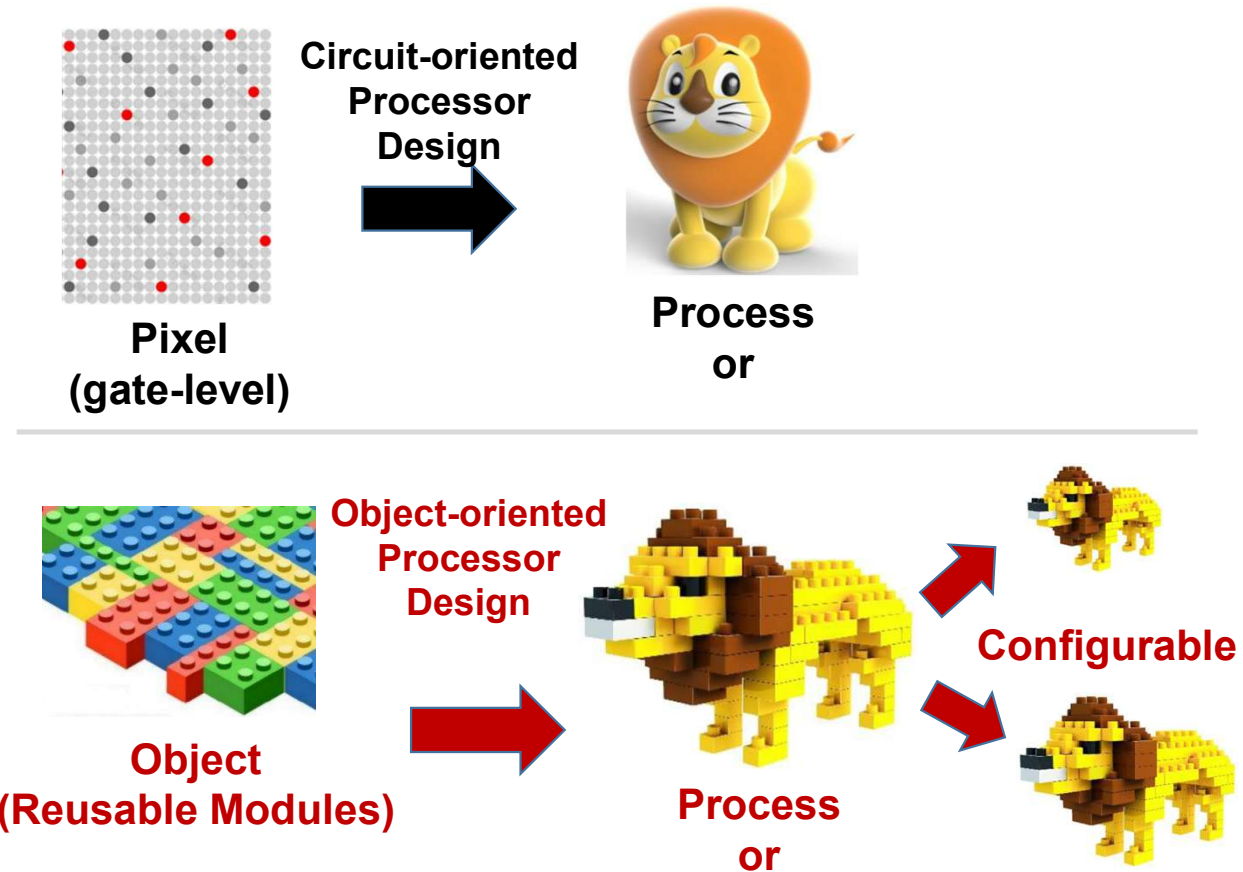
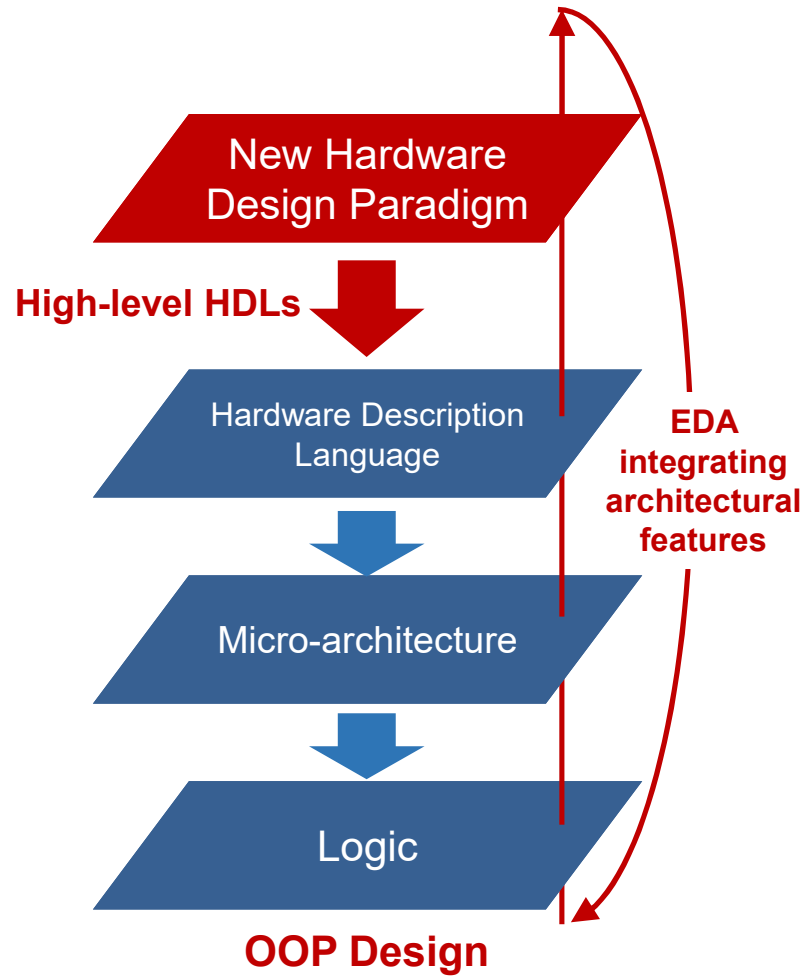


"BeiHai 100"
Labeled RISC-V Chip

- RV64GC Instruction Set
- In-order 9-stage pipeline
- Custom in-package coherent interconnect technology
- 8-core / 2MB L2 Cache
- ChipLink front-end bus
- **1.2GHz @ 28nm**
- Wafer out/WB BGA package
- Up to 32GB DDR4 memory support
- 2x Gigabit Ethernet
- 1x PCIe3.0 RC x4



New HDL → New Design Paradigm





What's Missing in Agile Hardware Design? Verification!

What's Missing in Agile Hardware Design? Verification!

Babak Falsafi, *Fellow, ACM, IEEE*

Parallel Systems Architecture Laboratory, Institute of Computer and Communication Sciences, School of Computer and Communication Sciences, Ecole polytechnique fédérale de Lausanne CH-1015 Lausanne

E-mail: babak.falsafi@epfl.ch

Agile hardware design is an approach to developing hardware systems that draws inspiration from the principles and practices of agile software development. It emphasizes collaboration, flexibility, iterative development, and quick adaptation to changing requirements. In agile hardware design, the focus is on delivering functional hardware systems in shorter development cycles while maintaining high-quality and customer satisfaction.

In particular, agile hardware design is of great interest in the open-source hardware community. Open-source hardware development — such as RISC-V — is at the forefront of initiatives to democratize hardware and drive innovation in chip design forward. Agile design is instrumental for the RISC-V community because it supports rapid iteration, accommodates the evolving RISC-V standard and the addition of custom extensions, improves community collaboration and time-to-market, and addresses the design challenges associated with complex architectural features.

Among significant innovations based on agile hardware design is the recently announced XIANGSHAN RISC-V core which is currently the highest performing RISC-V out-of-order microprocessor core with single-thread performance exceeding both existing RISC-V cores and a state-of-the-art ARM core, Cortex-A76. The creators of this platform have published their agile design methodology in a flagship computer architecture venue, MICRO, with a paper that has been selected through peer review to be among the best dozen papers in all of computer architecture in one year for publication in IEEE Micro Top Picks.

A key contributor to this breakthrough has been integrating hardware verification into the agile methodology. Hardware verification is crucial in designing digital platforms, as it ensures that semiconductor chips operate correctly and reliably according to the architecture specifications. Verification guarantees compliance with standards, and helps detect and rectify design errors, validate system-level functionality, optimize performance and power consumption, and enhance hardware reliability and safety. It plays a fundamental role in creating robust and dependable CPUs that meet the requirements of various applications and workloads.

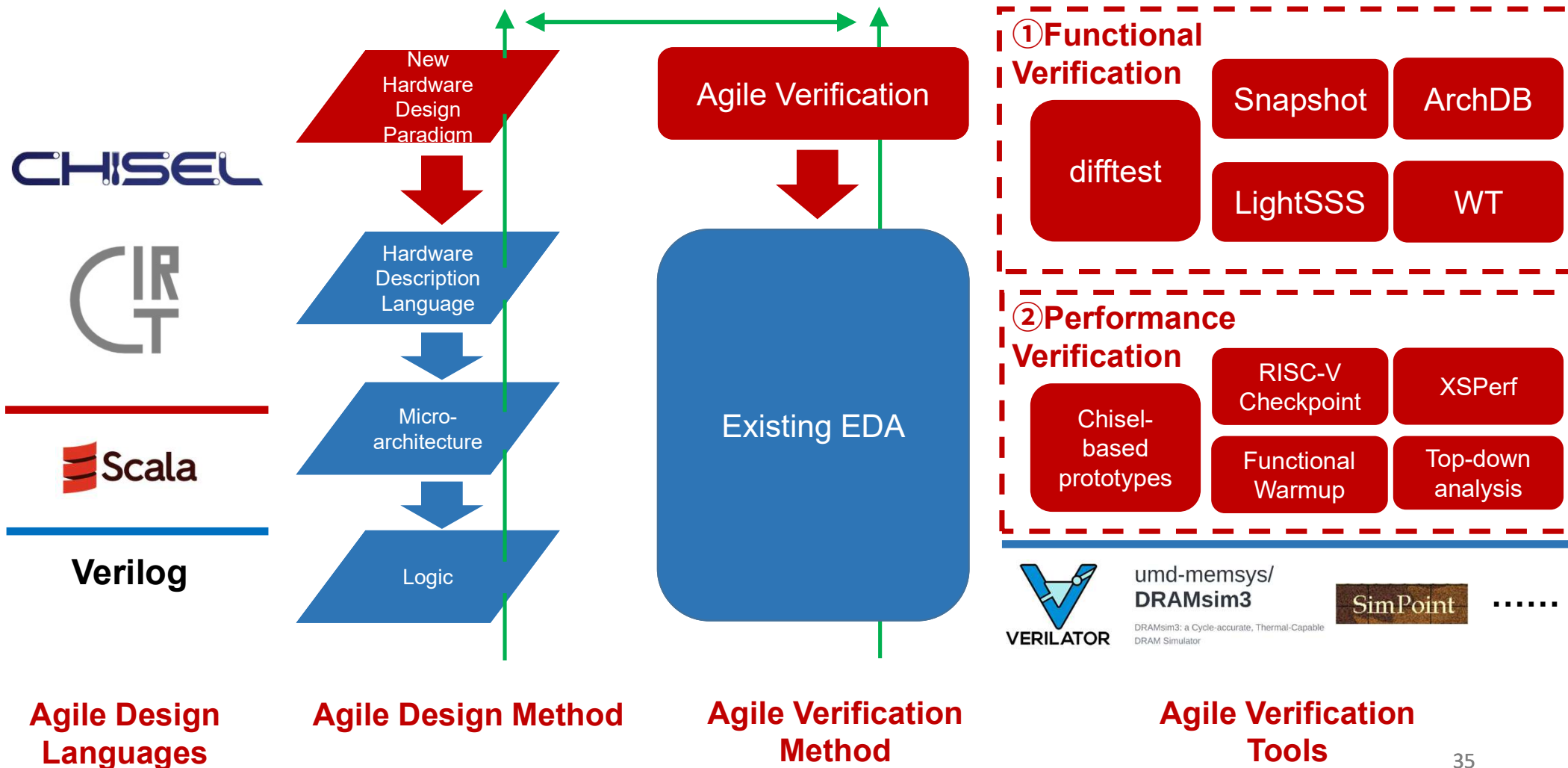


Babak Falsafi

Professor, EPFL

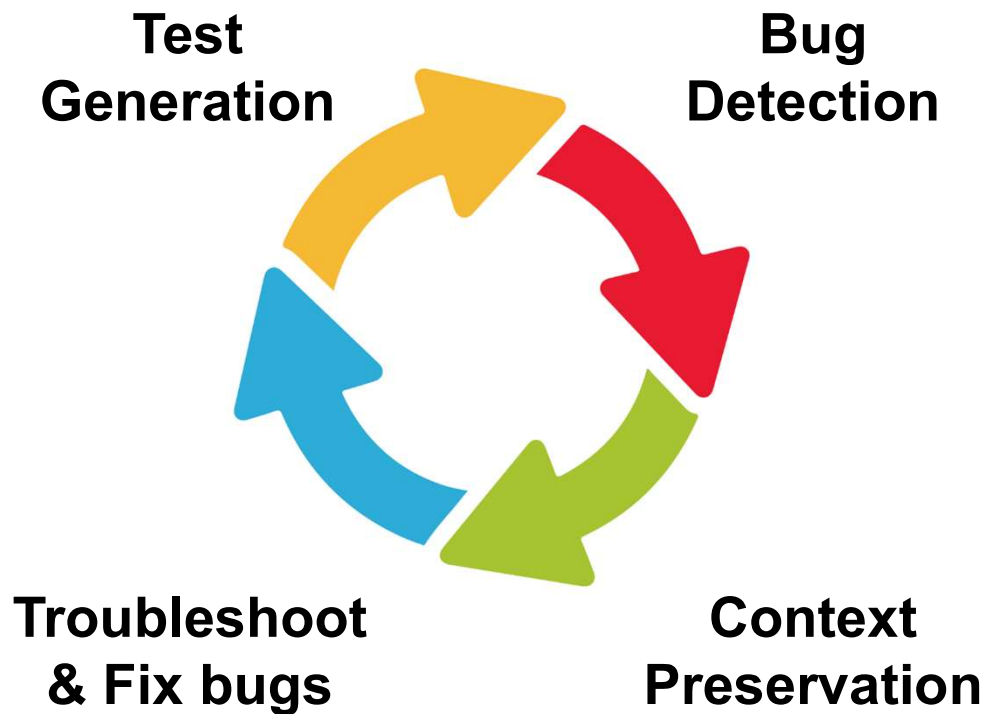
ACM/IEEE Fellow

Agile Verification is Challenging

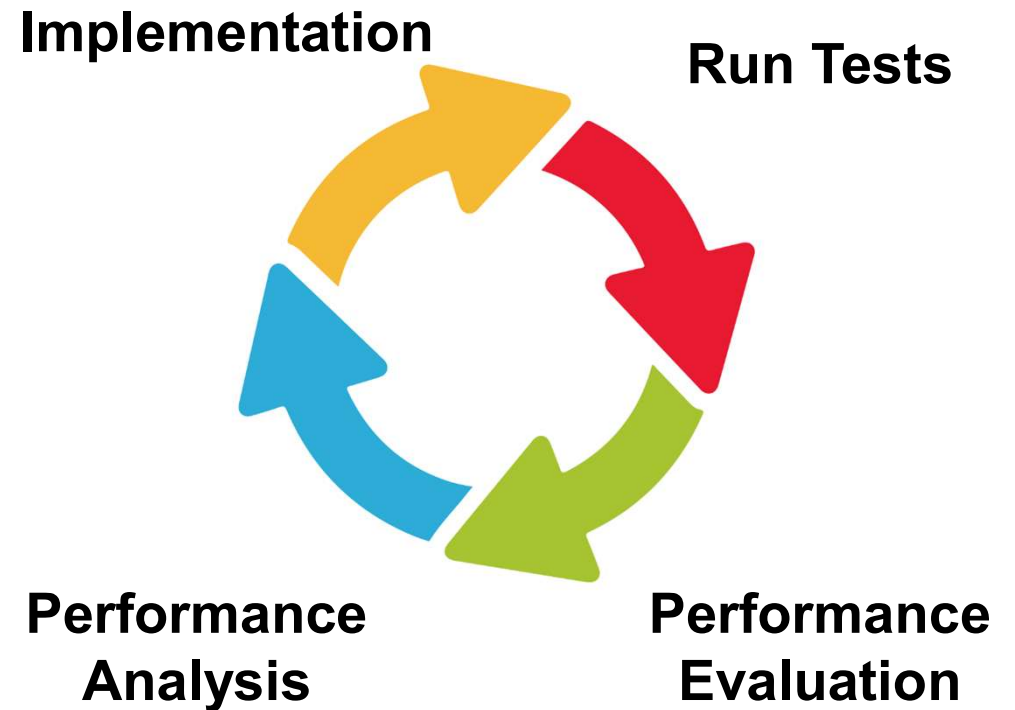


MinJie Loop

Functional Verification Loop

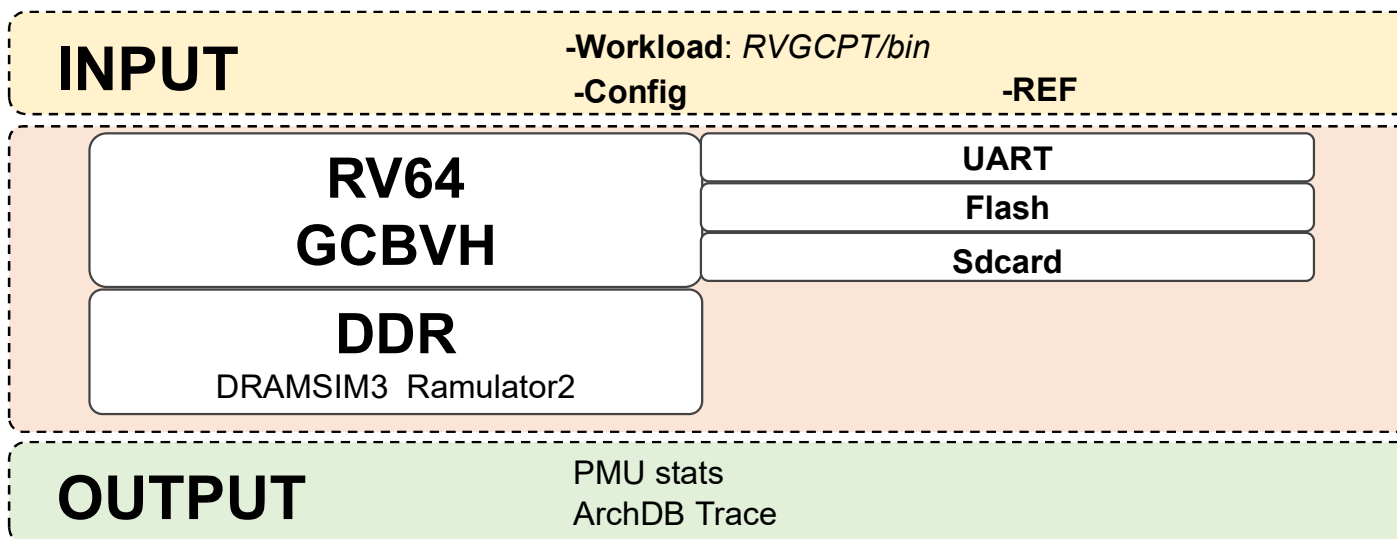
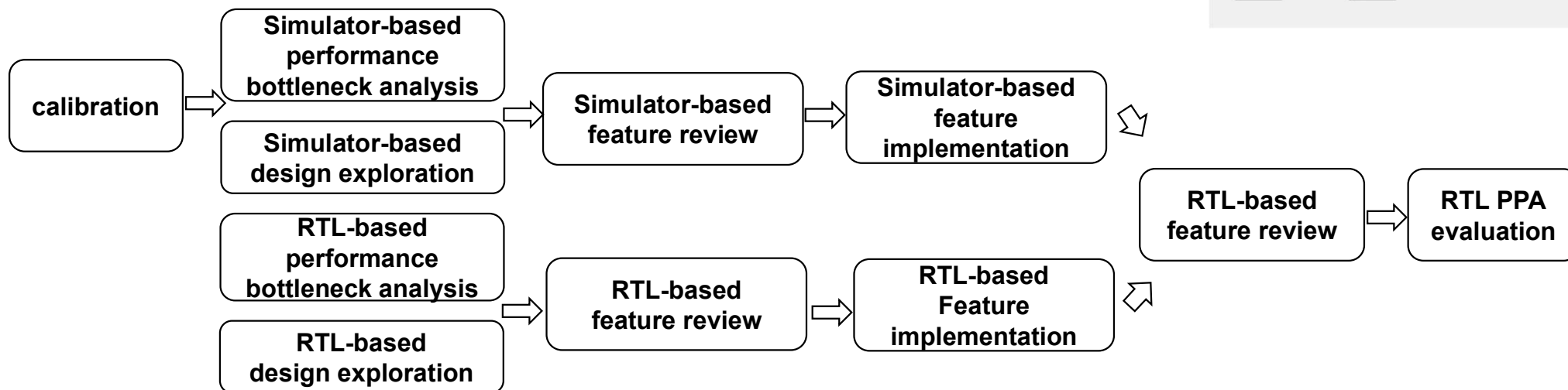


Performance Optimization Loop





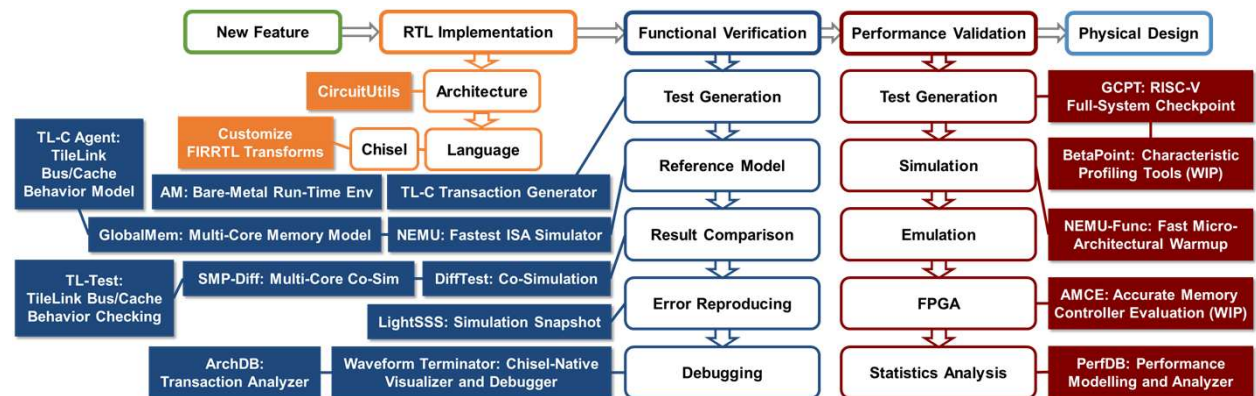
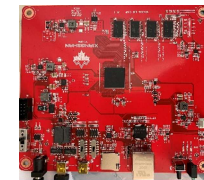
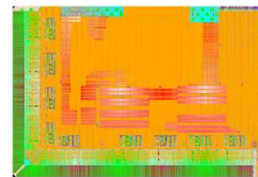
XS-Gem5: Calibrated Gem5 Simulator





MinJie: Open & Agile Verification Toolchain

- Infrastructure is the key outcome of the XiangShan Project
- Open source to benefit both academia and industry





Advanced Agile Chip Design Methodology

- New method and toolchain for agile chip design
- Developed more than 17 new tools to solve agile verification problems
- **MICRO 2022 → IEEE Micro Top Picks**

2022 55th IEEE/ACM International Symposium on Microarchitecture (MICRO)



Towards Developing High Performance RISC-V Processors Using Agile Methodology

Yinan Xu[†], Zihao Yu^{*}, Dan Tang[‡], Guokai Chen^{*†}, Lu Chen^{*†}, Lingrui Gou^{*†}, Yue Jin^{*†}, Qianruo Li^{*†}, Xin Li^{*†}, Zuojun Li^{*†}, Jiawei Lin^{*†}, Tong Liu^{*}, Zhigang Liu^{*}, Jiazhan Tan^{*}, Huaqiang Wang^{*†}, Huizhe Wang^{*†}, Kaifan Wang^{*†}, Chuanqi Zhang^{*†}, Fawang Zhang^{||}, Linjuan Zhang^{*†}, Zifei Zhang^{*†}, Yangyang Zhao^{*}, Yaoyang Zhou^{*†}, Yike Zhou^{*}, Jiangrui Zou^{||}, Ye Cai^{||}, Dandan Huan[§], Zusong Li[§], Jiye Zhao[§], Zihao Chen[§], Wei He[§], Qiyuan Quan[§], Xingwu Liu^{**}, Sa Wang^{*†}, Kan Shi^{*}, Ninghui Sun^{*†} and Yungang Bao^{*†}

^{*}State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, China

[†]University of Chinese Academy of Sciences, China

[‡]Beijing Institute of Open Source Chip, China

[§]Peng Cheng Laboratory, China

^{||}Beijing VCore Technology Co., Ltd., China

^{||}Shenzhen University, China

^{**}Dalian University of Technology, China

THEME ARTICLE: TOP PICKS FROM THE 2022 COMPUTER ARCHITECTURE CONFERENCES

Toward Developing High-Performance RISC-V Processors Using Agile Methodology

Yinan Xu[✉] and Zihao Yu[✉], State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, 100190, China

Dan Tang[✉], Beijing Institute of Open Source Chip, Beijing, 100080, China

Ye Cai[✉], Shenzhen University, Shenzhen, 518060, China

Dandan Huan[✉], Beijing VCore Technology, Beijing, 100190, China

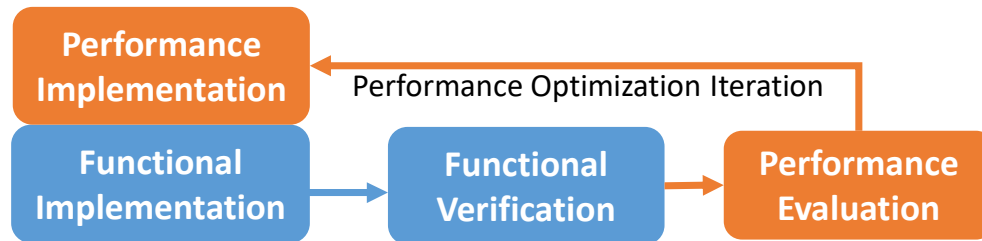
Wei He[✉], Peng Cheng Laboratory, Shenzhen, 518060, China

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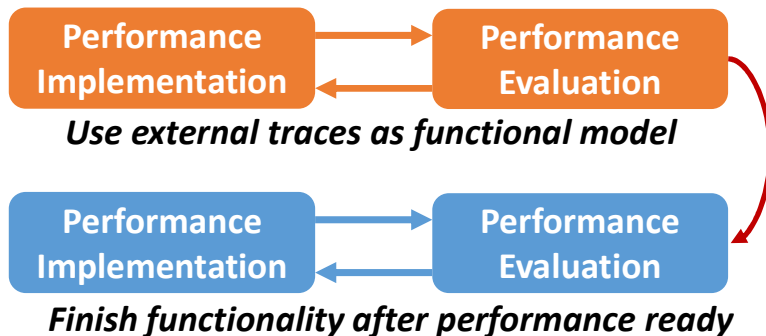


TraceRTL

Traditional Flow Pains: functionality first



TraceRTL Flow Gains: Performance first



Advantages

Extreme Efficiency

Turn RTL to “performance-first” model, shortening the prototyping cycle.

Ultra-High Accuracy

Inherit RTL details and mitigate trace-driven performance errors

Cross-ISA Compatibility

Simplify binary translation and evaluate programs in other ISAs

Oral Presentation at **Feb 3** in **Room C3.3**
Welcome to attend!



The book cover is light purple with a white border. At the top, the title "ISA Spec." is written in large, bold, blue sans-serif font. In the center, the subtitle "The RISC-V Instruction Set Manual" is in a smaller black font, followed by "Volume I: User-Level ISA" and "Document Version 2.2" in an even smaller font. Below this, the editors' names "Editores: Andrew Waterman¹, Kristi Asanović^{2,*}" are listed, followed by their affiliations: "¹SiFive Inc." and "²CS Division, EECS Department, University of California, Berkeley". At the bottom, the email addresses "as@eecs11100.berkeley.edu, kristi@eecs11100.berkeley.edu" and the date "May 7, 2017" are provided. At the very bottom, there is a yellow banner with a red circle containing the number "1" and the text "Open ISA" in large, bold, red font.

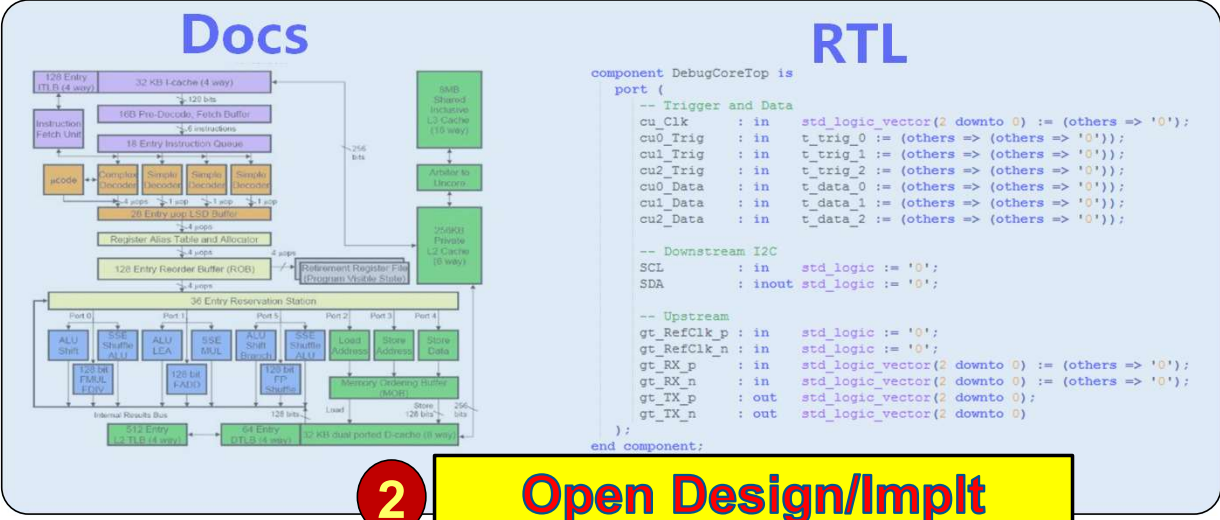
ISA Spec.

The RISC-V Instruction Set Manual
Volume I: User-Level ISA
Document Version 2.2

Editores: Andrew Waterman¹, Kristi Asanović^{2,*}
¹SiFive Inc.
²CS Division, EECS Department, University of California, Berkeley
as@eecs11100.berkeley.edu, kristi@eecs11100.berkeley.edu
May 7, 2017

1 Open ISA

1 Open ISA



2 Open Design/Implt

```

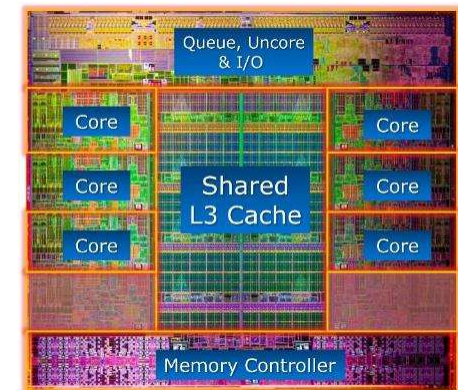
component DebugCoreTop is
port (
    -- Trigger and Data
    cu0_Clk      : in    std_logic_vector(2 downto 0) := (others => '0');
    cu0_Trig     : in    t_trig_0 := (others => (others => '0'));
    cu1_Trig     : in    t_trig_1 := (others => (others => '0'));
    cu2_Trig     : in    t_trig_2 := (others => (others => '0'));
    cu0_Data     : in    t_data_0 := (others => (others => '0'));
    cu1_Data     : in    t_data_1 := (others => (others => '0'));
    cu2_Data     : in    t_data_2 := (others => (others => '0'));

    -- Downstream I2C
    SCL          : in    std_logic := '0';
    SDA          : inout std_logic := '0';

    -- Upstream
    gt_RefClk_p  : in    std_logic := '0';
    gt_RefClk_n  : in    std_logic := '0';
    gt_RX_p      : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_RX_n      : in    std_logic_vector(2 downto 0) := (others => '0');
    gt_TX_p      : out   std_logic_vector(2 downto 0);
    gt_TX_n      : out   std_logic_vector(2 downto 0)
);
end component;

```

Layout



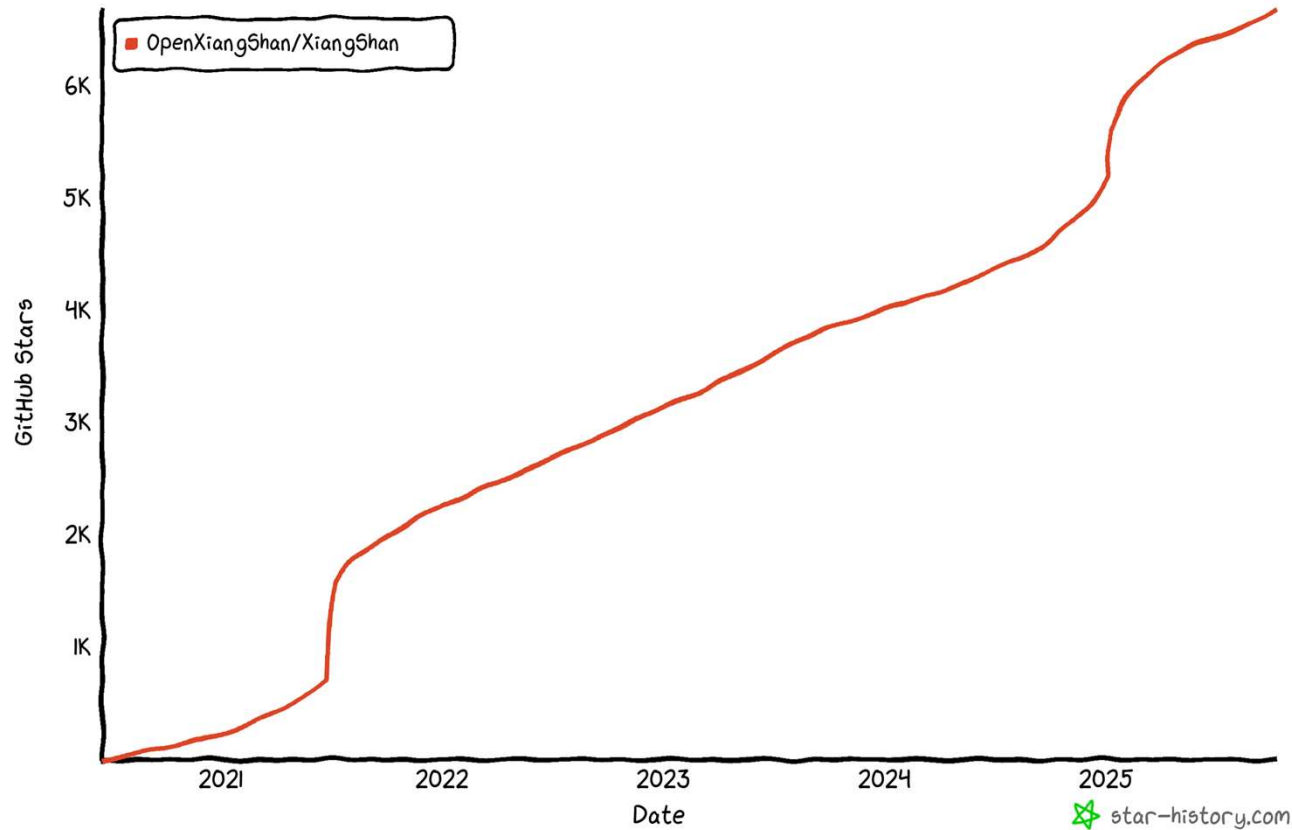
Prosperous Open-Source Community



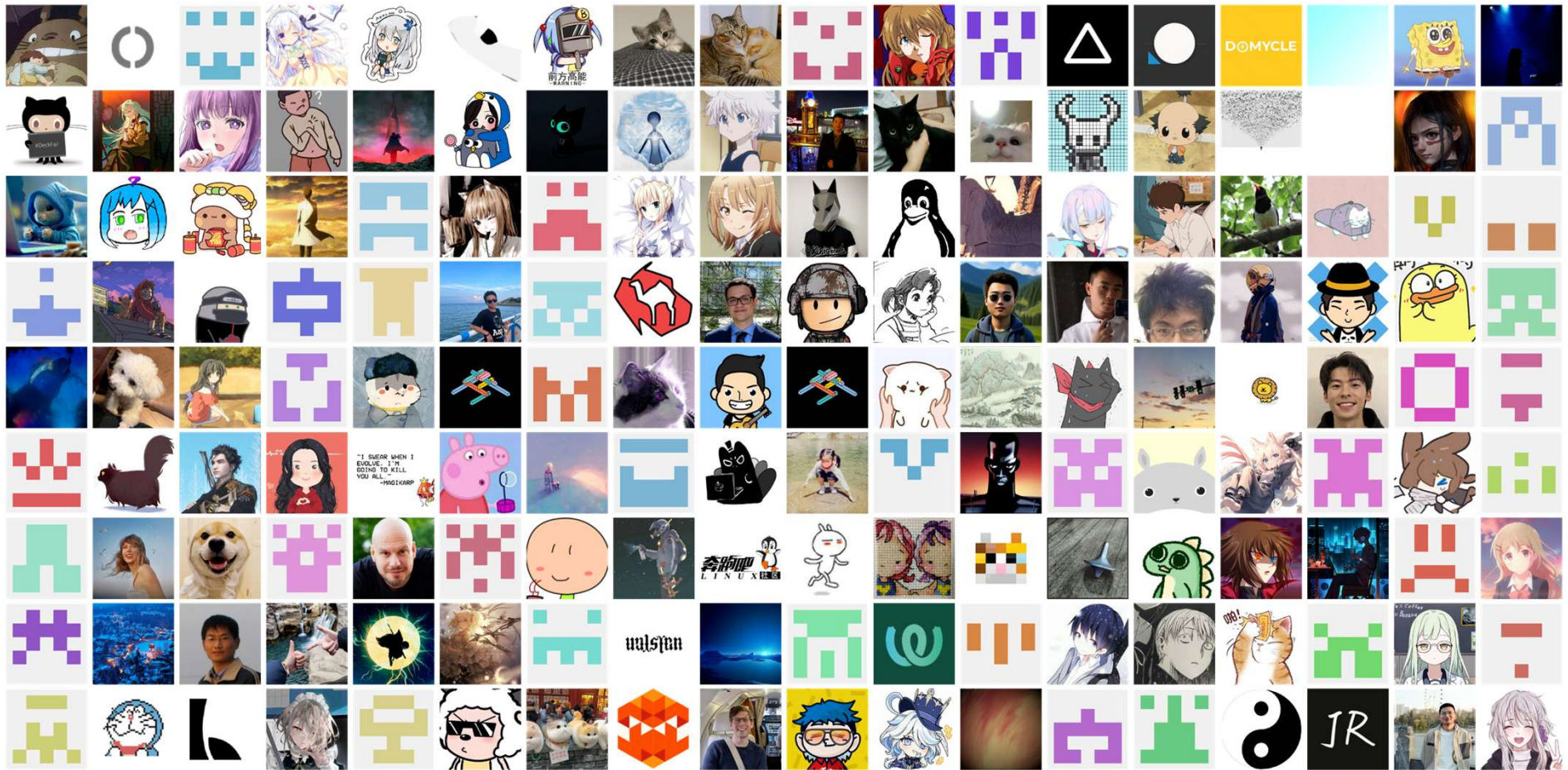
One of the Most Popular Open-Source Chip Projects

- **GitHub: >6800 Stars, > 850 Forks**

 Star History



Data collected on Oct 10, 2025



44

XiangShan Open-Source Community Conference



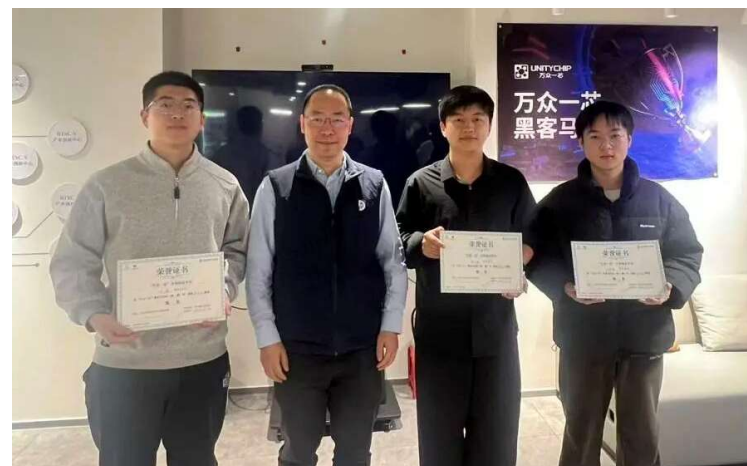
RISC-V Summit China 2024



RISC-V Summit China 2025

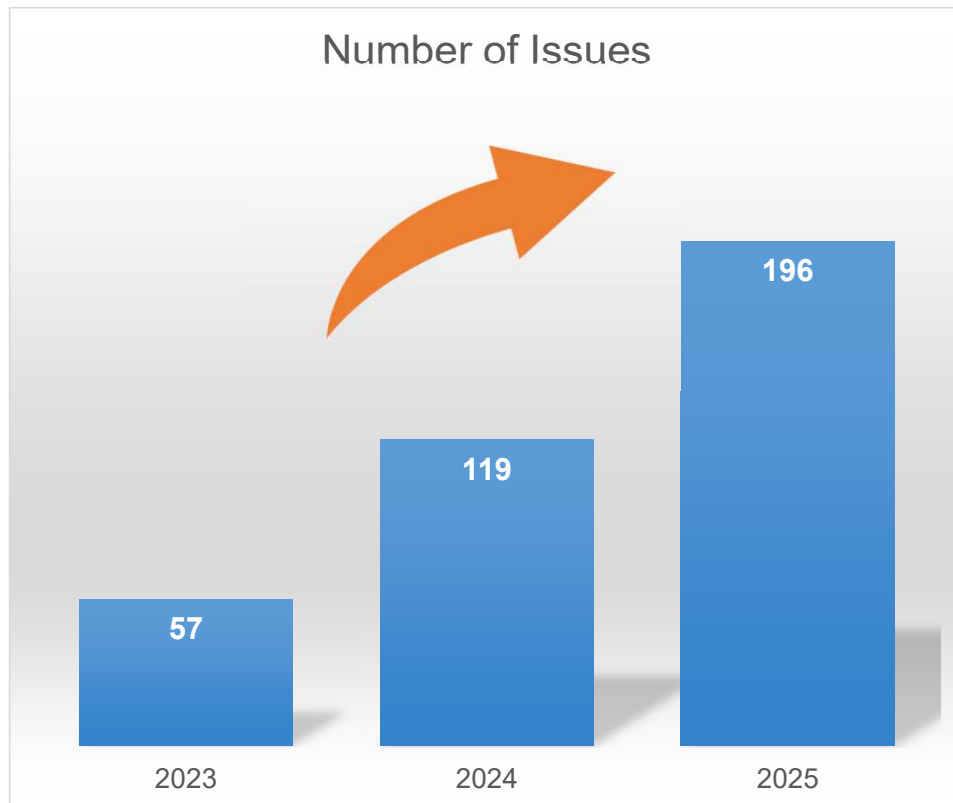


RISC-V Summit Europe 2025



UnityChip Open-Source Chip Hackathon

GitHub Issue & Discussion



- Time to First Response for Issues
 - Average: 25 hours
 - Median: 13 hours
- free to create a new issue on GitHub



Acknowledgments in XiangShan

- **Acknowledgments in XiangShan**

- A list outlines 32 techniques used in the XiangShan RTL codes
- <https://docs.xiangshan.cc/acknowledgments/>

- **Biweekly Report in English**

- our recent progress and performance data
- <https://docs.xiangshan.cc/zh-cn/latest/blog/category/biweekly-en/>

- **User Guide**

- For Software Developers and Hardware Integrators
- <https://docs.xiangshan.cc/projects/user-guide/en/latest/>

- **Design Document**

- Detailed Docs on Microarchitecture and Modules
- <https://docs.xiangshan.cc/projects/design/en/latest/>

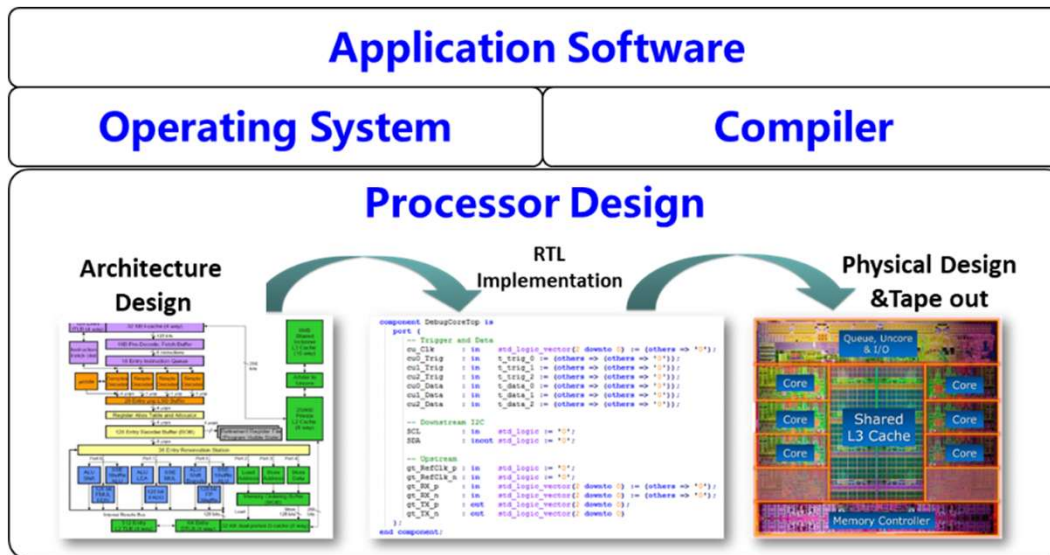


The One Student One Chip (OSOC) Initiative



OSOC Docs

- **Learning-by-Doing:** Teach undergraduates to build real chips
- Build the full stack of the computer system, **from HW to SW**
- Involves thousands of students in China and extends its impact overseas



The first microprocessor designed in Kazakhstan



UnityChip: Crowdsourcing Chip Verification

- **Features:** Involve software engineers

- Enable crowdsourcing
- Support multi-languages
- Be compatible with UVM

- **Effectiveness**

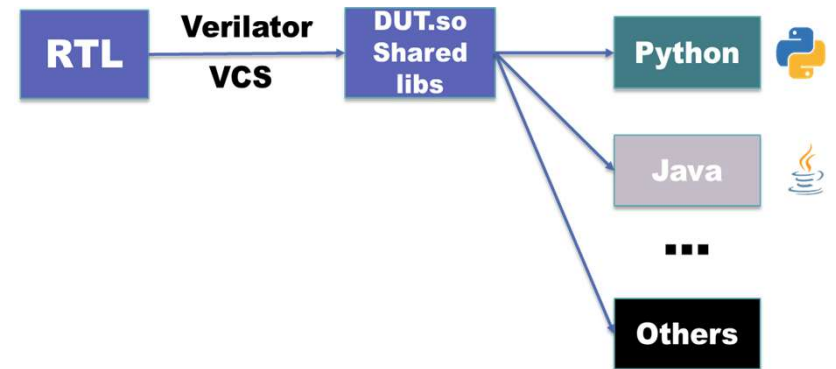
- 5 undergraduate students found **10 bugs** in 2 months

- **Easy-to-Use**

- Students not familiar with Linux and Python
- Learn to use tools in **5 days**
- Start adding test cases after another **10 days**

- **UnityChip Competition for XiangShan**

- Task: Unit Test for IFU, BPU, ITLB, etc.
- Scan the QR code for more information



GitHub Link

🏔️ UC-Agent: Automated AI Verification Agent

Focused on Chip Unit Test (UT)

- Accelerates the verification workflow by **delegating repetitive tasks to AI**, enabling engineers to focus on more challenging work
- Achieves **fully automated verification** for RTL code within approximately 5,000 lines, generating test cases, documentation, and reports
- Comprehensive verification workflow with highly configurable tools, **compatible with existing Code Agents**
- Supports leading domestic and international large language models with **high verification accuracy**



Open-source url: <https://github.com/XS-MLVP/UC-Agent>

XSCC: A High-Performance Compiler for RISC-V



XSCC is designed to serve

- ✓ Industrial-grade performance needs
- ✓ Open-source ecosystem development



Competitive performance



Engineering robustness

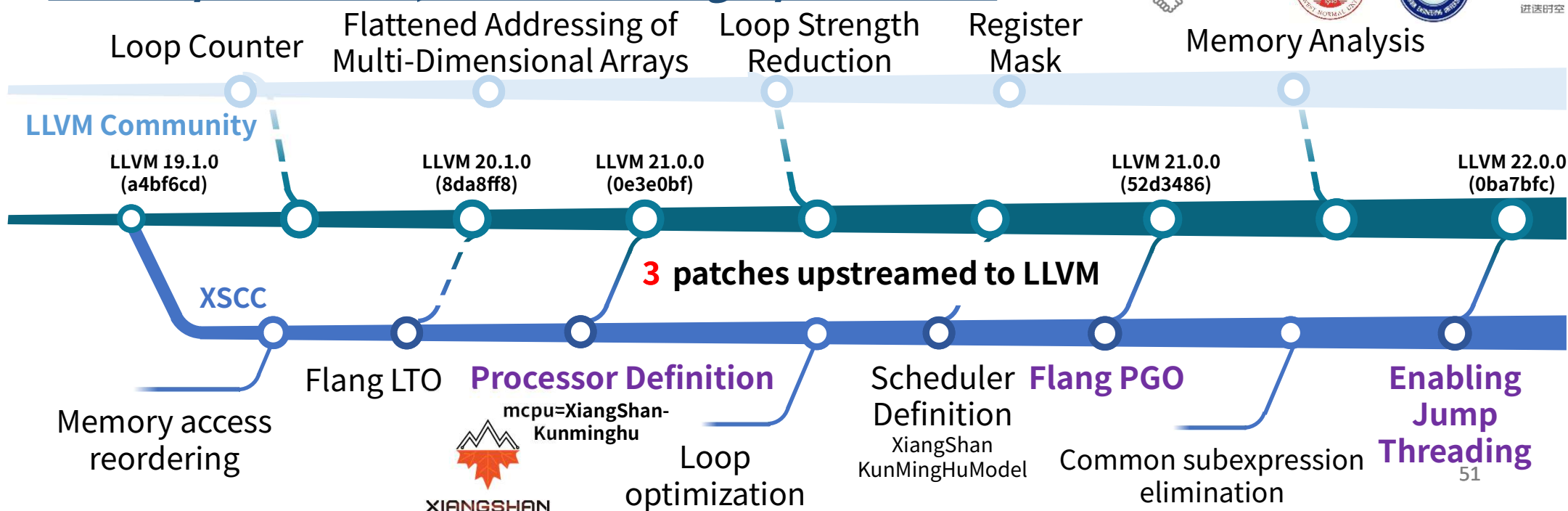


Deep integration with the RISC-V ecosystem

Core Goals

From Open Source, and Benefiting Open Source

Acknowledgements





Baiyang: High Performance Open-Source DDR-C IP

- Approaching the performance level of commercial IP
 - Integrated with XiangShan Kunminghu: **SPECCPU2006 score over 14/GHz**

Parameter configuration

- Supports different DDR modules and frequencies

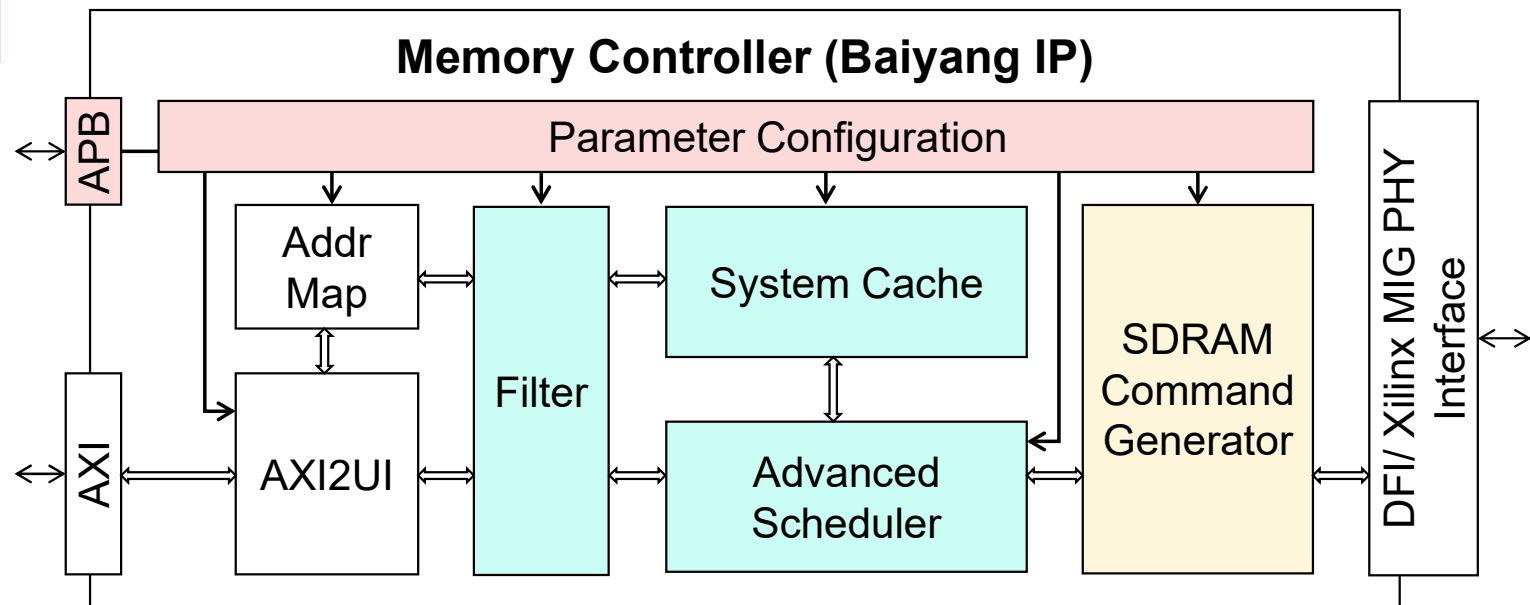
Controller Modules

- Decoupling DDR timing parameters from controller functionality

Rich functionality

- Advanced and configurable features: dynamic refresh, scheduling, cache, prefetch

<https://github.com/OpenXiangShan/YuQuan>



Together for a Shared Future

- Feel free to contact us through email or file issues on GitHub!
 - all@xiangshan.cc
 - <https://github.com/OpenXiangShan/XiangShan>



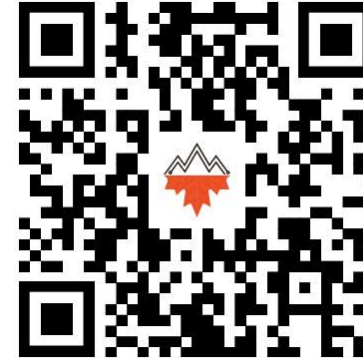
Homepage



Document



Biweekly
Report



User Guide



Design
Document